Physics Reference Notes

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1 Variational fermionic wave functions on lattices

1.1 Summary

1.2 Crash course on the origin of antisymmetric wave functions

This can be found in probably any introductory book on quantum mechanics, here I directly follow Schwabl, *Quantum Mechanics*, section 13.1.

The Heisenberg uncertainty relation implies that identical particles are indistinguishable. Consequently the probability amplitude $|\psi(x_1, \dots, x_N)|^2$ of a quantum many-body state is invariant under the exchange $x_i \leftrightarrow x_j$ of two indistinguishable particles. Define the exchange operator

$$P_{ij}\,\psi(x_1, \dots, x_i, \dots, x_j, \dots, x_N) \coloneqq \psi(x_1, \dots, x_j, \dots, x_i, \dots, x_N), \tag{1}$$

so that invariance of the amplitude gives

$$P_{ij}\,\psi(\boldsymbol{x}) = \lambda\,\psi(\boldsymbol{x}), \qquad |\lambda|^2 = 1. \tag{2}$$

Furthermore, exchanging the same two particles twice cannot change the wave function at all, hence

$$P_{ij}^2\,\psi(\boldsymbol{x}) = \lambda^2\psi(\boldsymbol{x}) = \psi(\boldsymbol{x}) \quad \Longrightarrow \quad \lambda^2 = 1, \text{ i.e. } \lambda = \pm 1. \tag{3}$$

Fermions are the particles for which $\lambda = -1$, i.e.

$$\boxed{\psi(\dots, x_i, \dots, x_j, \dots) = -\,\psi(\dots, x_j, \dots, x_i, \dots)} \tag{4}$$

and thus $\psi(\dots, x_i, \dots, x_i, \dots) = 0$.

1.3 Slater determinants

Here we loosely follow E. Koch, *Mean-Field Theory: Hartree-Fock and BCS* (Jülich 2024), section 1.

Let $\{\varphi_n\}$ be a complete, orthonormal single-particle basis,

$$\sum_n \varphi_n^*(x)\varphi_n(x') = \delta(x - x') \qquad \text{(complete)}, \tag{5}$$

$$\int dx\,\varphi_n^*(x)\varphi_m(x) = \delta_{nm} \qquad \text{(orthonormal)}. \tag{6}$$

An arbitrary N-particle function can then be expanded in its first argument as

$$a(x_1, \dots, x_N) = \sum_{n_1} a_{n_1}(x_2, \dots, x_N)\,\varphi_{n_1}(x_1), \qquad \text{with} \tag{7}$$

$$a_{n_1}(x_2, \dots, x_N) = \int dx_1'\,\varphi_{n_1}^*(x_1')\,a(x_1', x_2, \dots, x_N). \tag{8}$$

Iterating in all arguments, yields

$$\psi(x_1,\ldots,x_N) = \sum_{n_1,\ldots,n_N} a_{n_1,\ldots,n_N}\,\varphi_{n_1}(x_1)\cdots\varphi_{n_N}(x_N), \tag{9}$$

which is not antisymmetric yet. Now apply the antisymmetrizer $\mathcal{A}$, with $\mathrm{sgn}(P) = (-1)^P$:

$$\mathcal{A}\,\psi(x_1,\ldots,x_N) \overset{\text{[K (3)]}}{\equiv} \frac{1}{\sqrt{N!}}\sum_P (-1)^P\,\psi(x_{P(1)},\ldots,x_{P(N)}) \tag{10}$$

$$= \frac{1}{\sqrt{N!}}\sum_P (-1)^P \sum_{n_1,\ldots,n_N} a_{n_1,\ldots,n_N}\,\varphi_{n_1}(x_{P(1)})\cdots\varphi_{n_N}(x_{P(N)}) \tag{11}$$

$$= \sum_{n_1,\ldots,n_N} a_{n_1,\ldots,n_N}\,\Phi_{n_1,\ldots,n_N}(x_1,\ldots,x_N), \tag{12}$$

where we have used the *Slater determinant*

$$\Phi_{\alpha_1,\ldots,\alpha_N}(x_1,\ldots,x_N) \overset{\text{[K (4)]}}{\equiv} \frac{1}{\sqrt{N!}}\begin{vmatrix} \varphi_{\alpha_1}(x_1) & \varphi_{\alpha_2}(x_1) & \cdots & \varphi_{\alpha_N}(x_1) \\ \varphi_{\alpha_1}(x_2) & \varphi_{\alpha_2}(x_2) & \cdots & \varphi_{\alpha_N}(x_2) \\ \vdots & \vdots & \ddots & \vdots \\ \varphi_{\alpha_1}(x_N) & \varphi_{\alpha_2}(x_N) & \cdots & \varphi_{\alpha_N}(x_N) \end{vmatrix} \tag{13}$$

$$\overset{\text{Leibniz}}{=} \frac{1}{\sqrt{N!}}\sum_{P\in S_N} \mathrm{sgn}(P)\prod_{i=1}^{N}\varphi_{\alpha_i}\big(x_{P(i)}\big). \tag{14}$$

Hence **any** $\psi(x_1,\ldots,x_N)$ **can be expanded in Slater determinants.** Furthermore, it follows immediately that $\psi(\boldsymbol{x})$ vanishes if either $x_i = x_j$ or $\alpha_i = \alpha_j$ for $i \neq j$: **fermions cannot occupy the same position** x **or the same quantum state** α**.**

1.4 The variational principle

Following F. Becca & S. Sorella, *Quantum Monte Carlo Approaches for Correlated Systems*, section 1.4.

Formulate some variational state $|\psi_{\mathrm{var}}\rangle$ and consider

$$E_{\mathrm{var}} = \frac{\langle\psi_{\mathrm{var}}|H|\psi_{\mathrm{var}}\rangle}{\langle\psi_{\mathrm{var}}|\psi_{\mathrm{var}}\rangle} \qquad \text{[B\&S (1.38)]}. \tag{15}$$

Expanding in the energy eigenstates $|\varphi_i\rangle$,

$$|\psi_{\mathrm{var}}\rangle = \sum_i |\varphi_i\rangle\langle\varphi_i|\psi_{\mathrm{var}}\rangle = \sum_i a_i|\varphi_i\rangle \qquad \text{[B\&S (1.39)]}, \tag{16}$$

so that, using $\langle\varphi_i|\varphi_j\rangle = \delta_{ij}$,

$$E_{\mathrm{var}} = \Big(\sum_i |a_i|^2\Big)^{-1}\Big(\sum_i a_i|\varphi_i\rangle\Big)^\dagger H\Big(\sum_j a_j|\varphi_j\rangle\Big) = \Big(\sum_i |a_i|^2\Big)^{-1}\sum_i |a_i|^2 E_i. \tag{17}$$

Assume $E_i \leq E_j$ for $i < j$, i.e. E_0 is the ground-state energy. Then

$$E_{\text{var}} - E_0 = \Big(\underbrace{\sum_i |a_i|^2}_{>0} \Big)^{-1} \Big(\sum_{i \neq 0} \underbrace{|a_i|^2}_{\geq 0} \underbrace{(E_i - E_0)}_{\geq 0} \Big) \;\geq\; 0 \qquad \text{[B\&S (1.41)]}, \tag{18}$$

i.e. E_{var} **is an upper bound for the exact energy.** In practice: parametrize some $|\psi_{\text{var}}\rangle$ and minimize E_{var}.

1.5 Variational wave functions

1.5.1 Hartree-Fock wave functions

The Hartree-Fock (HF) approximation writes the many-body wave function as a product of single-particle orbitals,

$$\boxed{|\psi_{\text{HF}}\rangle = \prod_{\alpha=1}^{N_e} \phi_\alpha^\dagger |0\rangle, \qquad \phi_\alpha^\dagger = \sum_{i=1}^{N} \sum_\sigma W_{\sigma,\alpha,i}\, c_{i,\sigma}^\dagger, \quad \sigma = \uparrow, \downarrow} \tag{19}$$

with N lattice sites and N_e electrons. This leaves $2 \cdot N_e \cdot N$ parameters instead of the exponentially many amplitudes of a generic many-body state. The parameters are then optimized by minimizing

$$E_{\text{HF}} = \frac{\langle \psi_{\text{HF}} | H | \psi_{\text{HF}} \rangle}{\langle \psi_{\text{HF}} | \psi_{\text{HF}} \rangle}, \qquad \text{e.g.} \quad W_{\sigma,\alpha,i} \;\to\; W_{\sigma,\alpha,i} - \frac{\partial E_{\text{HF}}}{\partial W^*_{\sigma,\alpha,i}}\, \delta\tau \qquad \text{[B\&S (1.54)]}, \tag{20}$$

using Wirtinger derivatives (see section 12).

The HF wave function can equivalently be written in first quantization as **one Slater determinant**. Since it is a *product of single-particle orbitals*, it is mostly useful in the non-interacting case or at weak coupling. In the Hubbard model, turning up U makes the motion of one electron depend on where the other electrons are, i.e. it generates electron-electron correlation.

1.5.2 The Gutzwiller wave function

Gutzwiller's idea: describe the effect of the Hubbard U interaction by reducing the weight of configurations with multiply occupied sites. Start from some ground state $|\psi_0\rangle$ (e.g. non-interacting, a Slater determinant) and apply an operator P_g that suppresses such configurations:

$$|\psi_g\rangle = P_g |\psi_0\rangle \qquad \text{[B\&S (1.58)]} \tag{21}$$

$$\text{with} \qquad P_g = \exp\Big[-\frac{g}{2} \sum_{i=1}^{N} (n_i - n)^2 \Big] \qquad \text{[B\&S (1.59)]}, \tag{22}$$

where $n_i = \sum_\sigma n_{i\sigma}$ and $n = \frac{1}{N} \big\langle \sum_{i=1}^N n_i \big\rangle$ is the average density per site. Here g is the variational parameter ($g > 0$ for the repulsive Hubbard model).

Given some configuration $|x\rangle$, P_g counts how many sites within that state have higher or lower occupation than the average; for repulsive interactions these states are suppressed accordingly:

$$\langle x|\psi_g\rangle = P_g(x)\,\langle x|\psi_0\rangle, \qquad P_g(x) \leq 1 \qquad \text{[B\&S (1.60)]}. \tag{23}$$

Since $P_g(x) \geq 0$, it *does not* affect the sign structure.

In the Hubbard model a metal-insulator transition is expected at *finite* U/t for $n = 1$. The Gutzwiller wave function can only reproduce this for $g \to \infty$; otherwise there is always a non-zero probability for doublon-holon creation, and these could then move around freely and thus conduct.

Fully projected Gutzwiller wave function ($g \to \infty$).

Examine P_∞ for $n = 1$:

$$P_\infty = \lim_{g\to\infty} P_g = \lim_{g\to\infty} \exp\Big[-\frac{g}{2}\sum_{i=1}^{N}(\hat{n}_i - 1)^2\Big], \tag{24}$$

$$\begin{aligned} \Rightarrow \quad P_\infty|\psi\rangle &= \lim_{g\to\infty} \begin{cases} \exp\big(-\frac{g}{2}a\big), & \text{if } \exists \text{ site with a holon/doublon } (a>0) \\ \exp(0), & \text{else} \end{cases} |\psi\rangle \\ &= \begin{cases} 0 \\ |\psi\rangle \end{cases} \end{aligned} \tag{25}$$

Therefore we can rewrite P_∞ as (B&S (1.63))

$$P_\infty = \prod_i \big(n_{i,\uparrow} - n_{i,\downarrow}\big)^2. \tag{26}$$

If we want to allow empty sites, one has to set

$$P_\infty = \prod_i \big(1 - n_{i,\uparrow} n_{i,\downarrow}\big), \tag{27}$$

which is important for $n < 1$, since then holons and doublons do not come in pairs.

For $g = \infty$ no density fluctuations can occur, whereas in a true insulator density fluctuations do occur but produce no net current – e.g. double-occupancy fluctuations via virtual processes in the Hubbard model at large U[1].

1.5.3 The density-density Jastrow wave function

Since the Gutzwiller wave function does not correctly describe an insulator, the ansatz requires modification. We can, for instance, change the correlation term that is applied

[1] see *Lecture Notes on Correlation and Magnetism*, Patrik Fazekas, chapter 5, t-J model)

to $|\psi_0\rangle$ (or change $|\psi_0\rangle \rightsquigarrow |\psi_{U=\infty}\rangle$). The Jastrow factor does this by including long-range correlations:

$$\mathcal{J} = \exp\Big[-\frac{1}{2}\sum_{i,j} v_{ij}\,(n_i - n)(n_j - n)\Big] \qquad \text{[B\&S (1.65)]} \tag{28}$$

$$|\psi_{\mathcal{J}}\rangle = \mathcal{J}|\psi_0\rangle = |\psi_g\rangle, \quad \text{for } v_{ij} = \delta_{ij}\, g \qquad \text{[B\&S (1.64)]}. \tag{29}$$

For translationally invariant systems the variational parameters are $v_{ij} = v_{|i-j|}$, i.e. $\mathcal{O}(L)$ possible different distances.

The long-range correlations induced by $\mathcal{J}$ shall create a bound state between holons and doublons, possibly hindering conduction while still allowing local density fluctuations. It can be shown that the behaviour of v_{ij} resulting from an energy minimization is *not* enough to capture a fully gapped phase (power-law behaviour is present, but the Fourier transform of v_{ij} decays exponentially).

Jastrow-Slater wave function.

In the case $v_{ij} = v_{ji}$ the Jastrow factor is symmetric under particle exchange, so (in first quantization) it can be combined with a Slater determinant to obtain a suitable fermionic wave function, $\psi_{\mathrm{JS}}(\boldsymbol{x}) = \langle \boldsymbol{x}|\psi_{\mathrm{JS}}\rangle$:

$$\psi_{\mathrm{JS}}(\boldsymbol{r}_1, \ldots, \boldsymbol{r}_{N_e}) = \exp\Big[-\sum_{i<j} v_{ij}(n_i - n)(n_j - n)\Big] \cdot \psi_{\mathrm{HF}}(\boldsymbol{r}_1, \ldots, \boldsymbol{r}_{N_e}), \tag{30}$$

with $\psi_{\mathrm{HF}}(\boldsymbol{r}_1, \ldots, \boldsymbol{r}_{N_e}) = \det\{\phi_\alpha(\boldsymbol{r}_i)\}$ and $\{\phi_\alpha(\boldsymbol{r}_i)\}$ a set of one-particle orbitals (see B&S (1.68)).

1.5.4 The backflow wave function

As explained earlier, the set of Slater determinants forms a basis of the N-electron Hilbert space, so one can systematically improve HF by adding more Slater determinants, but this is computationally very demanding. Instead: **try to improve one single Slater determinant by making the orbitals configuration dependent**, which goes beyond mean field.

Original idea.

The movement of one particle "pushes other particles away" in order to avoid large overlap among them, i.e. a "counter flow" of all other particles – hence *backflow correlations*. One therefore constructs the Slater determinant with modified coordinates (continuum backflow),

$$\boldsymbol{r}_i^{\,b} = \boldsymbol{r}_i + \sum_{j\neq i} \eta\big(|\boldsymbol{r}_i - \boldsymbol{r}_j|\big)\big(\boldsymbol{r}_i - \boldsymbol{r}_j\big), \tag{31}$$

which also describes the effective displacement of the i-th particle due to the j-th one.

On a lattice.

For lattice models this effective displacement is much less useful, since all distances are discretized. Instead one changes the one-particle orbitals:

$$\phi_k\big(\boldsymbol{r}^{\,b}_{i,\sigma}\big) \;\simeq\; \phi^b_k\big(\boldsymbol{r}_{i,\sigma}\big) \equiv \phi_k\big(\boldsymbol{r}_{i,\sigma}\big) + \sum_j c_{ij}[x]\,\phi_k\big(\boldsymbol{r}_{j,\sigma}\big) \tag{32}$$

where $\phi_k(\boldsymbol{r}_{i,\sigma}) = \langle 0|c_{i,\sigma}|\phi_k\rangle$, the $|\phi_k\rangle$ are eigenstates of the mean-field Hamiltonian, and $|x\rangle$ is the many-body configuration of the system.

This is essentially a *dynamic linear combination of orbitals*: dynamic, since it *depends on the electron configuration*. In the Hubbard model at large U one can hybridize the orbitals as follows:

$$\phi^b_k(\boldsymbol{r}_{i,\sigma}) = \varepsilon\,\phi_k(\boldsymbol{r}_{i,\sigma}) + \eta \sum_j t_{ij}\left(D_i H_j\right)\phi_k(\boldsymbol{r}_{j,\sigma}) \tag{33}$$

i.e. hybridize the orbitals at i and j if there is a doublon at i and a holon at j, such that the electron can minimize energy by going to the empty site[2].

Summary.

Backflow correlations improve a single Slater determinant by replacing fixed single-particle orbitals with **configuration-dependent orbitals**, effectively introducing **many-body corrections** through a determinant whose elements depend on the full configuration. Combining this with a Jastrow factor gives the Slater-Jastrow-backflow wave function.

1.6 Fermionic systems with determinants

In this subsection we review section 5.6 of the book *Quantum Monte Carlo Approaches for Correlated Systems* by Federico Becca and Sandro Sorella. The presentation closely follows Becca and Sorella, with additional details included for clarity.

Throughout this section, we assume a lattice that contains L sites and each site can contain one spin-up and one spin-down orbital. Therefore, we have $2L$ available orbitals that can be occupied by particles. We use one collective index that contains the site and the spin, by setting

$$\hat{c}_{i,\uparrow} \equiv \hat{d}_i \quad \text{and} \quad \hat{c}_{i,\downarrow} \equiv \hat{d}_{i+L}. \tag{34}$$

This corresponds to the fermionic-mode ordering $(1\uparrow, 2\uparrow, \ldots, L\uparrow, 1\downarrow, 2\downarrow, \ldots, L\downarrow)$. Any fixed ordering is fine, but it must be used consistently throughout.

[2] cf. *"Role of backflow correlations for the non-magnetic phase of the t-t′ Hubbard model"* by Tocchio and Sorella, Eqs. (2)-(4)

Any number-conserving quadratic fermionic Hamiltonian can be diagonalized:

$$\hat{H}_0 = \sum_{I,J=1}^{2L} t_{I,J} d_I^\dagger \hat{d}_J \tag{35}$$

$$= \boldsymbol{d}^\dagger \boldsymbol{T} \boldsymbol{d} = \boldsymbol{d}^\dagger \boldsymbol{U} \boldsymbol{U}^\dagger \boldsymbol{T} \boldsymbol{U} \boldsymbol{U}^\dagger \boldsymbol{d} \tag{36}$$

$$= \boldsymbol{\phi}^\dagger \boldsymbol{E} \boldsymbol{\phi} = \sum_{\alpha=1}^{2L} \varepsilon_\alpha \hat{\Phi}_\alpha^\dagger \hat{\Phi}_\alpha, \tag{37}$$

$$\text{with} \quad \phi_\alpha^\dagger = \sum_I U_{I,\alpha} d_I^\dagger. \tag{38}$$

The ground state is thus obtained by occupying the N_e lowest-energy single-particle orbitals:

$$|\Phi_0\rangle = \prod_{\alpha=1}^{N_e} \hat{\Phi}_\alpha^\dagger |0\rangle = \prod_{\alpha=1}^{N_e} \left(\sum_{I=1}^{2L} U_{I\alpha} \hat{d}_I^\dagger \right) |0\rangle . \tag{39}$$

The matrix U is in general a $2L \times 2L$ unitary matrix, where we only need the first N_e columns to construct the ground state, as can be seen above.

A general occupation-number configuration can be written as

$$|x\rangle = \hat{d}_{R_1}^\dagger \hat{d}_{R_2}^\dagger \cdots \hat{d}_{R_{N_e}}^\dagger |0\rangle , \tag{40}$$

where $R_1, \ldots, R_{N_e}$ are the occupied orbitals in the configuration x, with $R_j = 1, \ldots 2L$. This is of course a redundant labeling, since for example $c_1^\dagger c_2^\dagger |0\rangle = -c_2^\dagger c_1^\dagger |0\rangle$. But to describe arbitrary configurations, that can occur from applying the Hamiltonian to another arbitrary configuration, they are convenient. The resulting operator string need not be in canonical order and it then must be reordered, producing the appropriate fermionic sign. Thus, different orderings may represent the same physical configuration up to a sign. Therefore, we have

$$\begin{aligned} \langle x|y\rangle &= \langle 0| \hat{d}_{R_{N_e}} \cdots \hat{d}_{R_1} \hat{d}_{Q_1}^\dagger \cdots \hat{d}_{Q_{N_e}}^\dagger |0\rangle \\ &= \begin{cases} 0, & \text{iff } \{R_1, \ldots, R_{N_e}\} \neq \{Q_1, \ldots, Q_{N_e}\} \\ \text{sign}(p), & \text{else, with } p\left((Q_1, \ldots, Q_{N_e})\right) = (R_1, \ldots, R_{N_e}), \end{cases} \end{aligned} \tag{41}$$

where $\{.\}$ denotes a mathematical set and $(.)$ an ordered tuple.

The overlap between a general configuration $|x\rangle$ and the ground state of the quadratic Hamiltonian is thus given as

$$\begin{aligned} \langle x|\Phi_0\rangle &= \langle 0| \hat{d}_{R_{N_e}} \cdots \hat{d}_{R_1} \prod_{\alpha=1}^{N_e} \left(\sum_{i=1}^{2L} U_{i\alpha} \hat{d}_i^\dagger \right) |0\rangle \\ &= \langle 0| \hat{d}_{R_{N_e}} \cdots \hat{d}_{R_1} \left[\sum_{p \in P} \text{sign}(p) \prod_{\alpha=1}^{N_e} U_{p\{Q_j\},\alpha} \right] \hat{d}_{Q_1}^\dagger \cdots \hat{d}_{Q_{N_e}}^\dagger |0\rangle , \end{aligned} \tag{42}$$

with P containing all possible permutations of the Q_j. These have now to be restricted on all valid permutations of the R_j within x (see eq. (41)), which yields

$$\langle x|\Phi_0\rangle = \langle 0|\, \hat{d}_{R_{N_e}} \cdots \hat{d}_{R_1} \left[\sum_{p\in P} \mathrm{sign}(p) \prod_{\alpha=1}^{N_e} U_{R_{p\{Q_j\}},\alpha}\right] \hat{d}^\dagger_{R_1} \cdots \hat{d}^\dagger_{R_{N_e}} |0\rangle \tag{43}$$

$$= \sum_{p\in P} \mathrm{sign}(p) \prod_{\alpha=1}^{N_e} U_{R_{p\{Q_j\}},\alpha} \tag{44}$$

$$= \det\left[U_{R_i,\alpha}\right]_{\substack{i=1,\ldots,N_e \\ \alpha=1,\ldots,N_e}}, \tag{45}$$

where we used the Leibniz formula in the last step.

In other words, although U is a $2L \times 2L$ matrix, only the N_e columns corresponding to the occupied lowest-energy orbitals and the N_e rows corresponding to the occupied orbitals R_i of the configuration x are needed:

$$U_x = \begin{pmatrix} U_{R_1,1} & \cdots & U_{R_1,N_e} \\ \vdots & \ddots & \vdots \\ U_{R_{N_e},1} & \cdots & U_{R_{N_e},N_e} \end{pmatrix} \tag{46}$$

and hence

$$\langle x|\Phi_0\rangle = \det(U_x). \tag{47}$$

This construction can also be used for BCS-type Hamiltonians (see end of section 5.6.1 of the book by Becca and Sorella).

Variational Monte Carlo

Let's say we have some general fermionic Hamiltonian $\hat{H}$, which might not be quadratic and we choose the Slater determinant as our variational ansatz $\langle x|\Phi_U\rangle = \det(U_x)$ with the entries of the full U as the variational parameters. The variational Monte Carlo (VMC) workflow then roughly looks like

1. Sample fermionic occupation configurations $x = (R_1, \ldots, R_{N_e})$ following the distribution resulting from $\Phi_U(x) = \det(U_x)$.
2. Evaluate Hamiltonian matrix elements $\langle x|\, \hat{H}\, |x'\rangle$ and estimate the local energy $e_L(x) = \sum_{x'} \langle x|\, \hat{H}\, |x'\rangle \frac{\langle x'|\Phi_U\rangle}{\langle x|\Phi_U\rangle}$.
3. Compute the gradient of the energy $\langle E\rangle \approx \frac{1}{N}\sum_{n=1}^{N} e_L(x_n)$ and optimize the variational parameters with it.
4. Repeat until convergence.

The Jordan-Wigner mapping

The Jordan-Wigner mapping (JWM) does not change anything in the derivation above since it is only a *rewriting of the operators* (and besides doing the above derivation with the JWM

would probably be much more tedious). But it might come in handy when computing matrix elements such as $\langle x|\,\hat{H}\,|x'\rangle$. To see what changes in the VMC workflow described above, we only have to look at how the form of the ansatz $\Phi_U(x)$ might change, since we know that the fermionic and spin configurations map onto each other (see section 2). But again, since the JWM is simply a rewriting of operators, the overlap of two states stays exactly the same and therefore the determinant ansatz remains. **Hence, the only significant change in the VMC workflow is that the Hamiltonian has to be rewritten but the ansatz remains unchanged.**

Computing matrix elements as for instance $\langle x|\,\hat{H}\,|x'\rangle$ involves commuting fermionic operators, which necessitates some book keeping for the resulting sign. This book keeping is exactly what is done by the Jordan-Wigner strings. Thus, in practice or numerically, we often use the Jordan-Wigner in order to circumvent having to do this book keeping by hand.

1.7 Parametrization via neural networks

Now let us have a look at how neural networks are used for variational states. There is no single resource that reviews the physics-motivated approaches like backflow, hidden fermions etc., so we mostly look at the corresponding papers.

1.7.1 Backflow transformations via neural networks

Following D. Luo & B.K. Clark, *Backflow Transformations via Neural Networks for Quantum Many-Body Wave Functions*.

Recall the definition of backflow introduced by Tocchio et al. in "Role of backflow correlations for the non-magnetic phase of the t-t' Hubbard model", Eq. (4):

$$\phi^b_{k\sigma}\big(r_{i,\sigma};\boldsymbol{x}\big) = \phi_{k\sigma}\big(r_{i,\sigma}\big) + \sum_j \eta_{ij,\sigma}\,\phi_{k\sigma}\big(r_{j,\sigma}\big), \tag{48}$$

$$\eta_{ij,\sigma} = t\,D_i H_j\,\Theta_{|i-j|,\sigma}, \qquad \text{(L\&C (6))} \tag{49}$$

with $D_i = n_{i,\uparrow}n_{i,\downarrow}$ and $H_i = (1-n_{i,\uparrow})(1-n_{i,\downarrow})$ counting doublons and holons.

The i,j-dependence of the hopping amplitudes t_{ij} is absorbed in the variational parameters $\Theta_{|i-j|,\sigma}$; e.g. for next-nearest-neighbour hopping one sets them to zero if $|i-j| \neq 1,2$.

Now we make the ansatz to parametrize the backflow part via a neural network:

$$\phi^b_{k\sigma}\big(r_{i,\sigma};\boldsymbol{x}\big) = \phi_{k\sigma}\big(r_{i,\sigma}\big) + a^{\mathrm{NN}}_{ki,\sigma}(\boldsymbol{x}) \qquad \text{(L\&C (5))} \tag{50}$$

The typical advantage of a NN parametrization is that neural networks are universal approximators. On top of that, one can show **explicitly how to construct the backflow with a NN with two hidden layers**:

Step 1: rewrite η as a ReLU.

Assuming $t\,\Theta_{|i-j|,\sigma} \geq 0$,

$$\eta_{ij,\sigma} = t\,D_i H_j\,\Theta_{|i-j|,\sigma} = t\,n_{i,\uparrow}n_{i,\downarrow}h_{j,\uparrow}h_{j,\downarrow}\,\Theta_{|i-j|,\sigma},$$

which is nonzero only if $n_{i,\uparrow} = n_{i,\downarrow} = h_{j,\uparrow} = h_{j,\downarrow} = 1$, and hence

$$\eta_{ij,\sigma} = \mathrm{ReLU}\Big[t\,\Theta_{|i-j|,\sigma}\big(n_{i,\uparrow} + n_{i,\downarrow} + h_{j,\uparrow} + h_{j,\downarrow} - 3\big)\Big].$$

Step 2: input layer.

The NN shall have the input layer

$$\big(\underbrace{\sigma_1, \ldots, \sigma_N}_{\text{spin up}}, \underbrace{\sigma_{N+1}, \ldots, \sigma_{2N}}_{\text{spin down}}\big), \qquad \text{with}$$

$$\sigma_i = 2n_{i,\uparrow} - 1, \quad \sigma_{i+N} = 2n_{i,\downarrow} - 1, \quad i = 1, \ldots, N.$$

Inverting the relation for σ_i and inserting it into $\eta_{ij,\sigma}$,

$$\eta_{ij,\sigma} = \mathrm{ReLU}\Big[\tfrac{1}{2}\,t\,\Theta_{|i-j|,\sigma}\big(\sigma_i + \sigma_{i+N} - \sigma_j - \sigma_{j+N} - 2\big)\Big],$$

with $i, j \in \{1, \ldots, N\}$ and $\sigma = \pm 1$.

Step 3: first hidden layer.

Use this to construct $\eta_{ij,\sigma}$ as the following NN (cf. fig. 1):

$$v_k = \sum_{\ell=1}^{2N} w_{k\ell}\,\sigma_\ell + b_k, \qquad \text{set} \quad k = \sigma \cdot (i + N \cdot j). \tag{51}$$

$$\Rightarrow \quad v_{\sigma\cdot(i+N\cdot j)} = \sum_{\ell=1}^{2N} w_{\sigma\cdot(i+N\cdot j),\ell}\,\sigma_\ell + b_{\sigma\cdot(i+N\cdot j)}, \tag{52}$$

with

$$b_{\sigma\cdot(i+N\cdot j)} = -t\,\Theta_{|i-j|,\sigma} \qquad \text{and} \qquad w_{\sigma\cdot(i+N\cdot j),\ell} = \frac{t}{2}\,\Theta_{|i-j|,\sigma} \begin{cases} 1 & \text{if } \ell \,\%\, N = i \\ -1 & \text{if } \ell \,\%\, N = j \\ 0 & \text{else} \end{cases} \tag{53}$$

such that $\eta_{ij,\sigma} = \mathrm{ReLU}\big(v_{\sigma\cdot(i+N\cdot j)}\big)$.

Step 4: second hidden layer.

Finally, the full backflow is obtained by adding another hidden layer which takes the $\eta_{ij,\sigma}$ as inputs and weights them by the single-particle orbitals $\phi_k(r_{j,\sigma})$. The activation function of this layer is simply the identity. If $t\,\Theta_{|i-j|,\sigma} < 0$, the minus sign can be absorbed into this layer, resulting in

$$a^{\mathrm{NN}}_{ki,\sigma}(\boldsymbol{r}) = \sum_j \underbrace{\eta_{ij,\sigma}}_{\text{inputs}}\,\underbrace{s_{ij,\sigma}\,\phi_{k\sigma}(r_{j,\sigma})}_{\text{weights}}, \qquad s_{ij,\sigma} = \mathrm{sgn}\big(t\,\Theta_{|i-j|,\sigma}\big) = \pm 1. \tag{54}$$

$\Rightarrow$ the backflow can be represented by a neural network.

In practice, one simply chooses some NN (RBM, MLP, CNN, ...) and lets backpropagation take care of the rest.

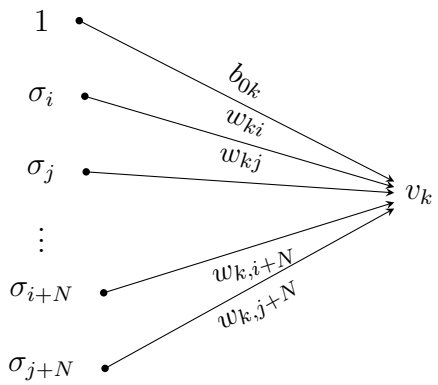


Figure 1: First hidden layer producing v_k, with $k = \sigma \cdot (i + N \cdot j)$.

1.8 References

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2 The Jordan-Wigner mapping for spinless Fermions

2.1 Summary

The Jordan-Wigner transformation (JWT) can be used to map a fermionic system to a system of spins. This is essentially done by rewriting

$$\text{fermion} = \text{string} \times \text{operator},$$

where the "string" is some product of operators $\hat{Z}_i = 1 - 2\hat{n}_i$ and the "operator" is found to obey spin commutation relations. Within this section, we only consider spinless fermions, but the JWT can also be applied on spinful systems.

The importance of the JWT can be seen, when examining how to compute some general matrix element $\langle x| \hat{A} |x'\rangle$, where the operator $\hat{A}$ is given as some product of fermionic creation and annihilation operators. To determine the matrix element, one has to commute operators and keep track of the resulting minus signs. This is exactly what the Jordan-Wigner mapping does for us.

Refer to section 1.6 to see what role the JWT plays in variational Monte Carlo.

2.2 Setup: the fermionic Hilbert space

Our fermionic Hilbert space is defined by (Nielsen):

i) operators $c_1, \ldots, c_L$ acting on some Hilbert space;

ii) these operators satisfying the canonical (anti-)commutation relations for fermions

$$\{c_i^{(+)}, c_j^{(+)}\} = 0 \qquad \& \qquad \{c_i, c_j^\dagger\} = \delta_{ij}\, \mathbb{1}. \tag{55}$$

Many things directly follow from this, such as a possible basis for the Hilbert space, namely the occupation number basis

$$|\boldsymbol{n}\rangle = |n_1, n_2, \ldots, n_L\rangle \equiv \left(\prod_{i=1}^{L} (c_i^\dagger)^{n_i} \right) |0\rangle\,, \tag{56}$$

where $\hat{n}_i |\boldsymbol{n}\rangle = c_i^\dagger c_i |\boldsymbol{n}\rangle = n_i |\boldsymbol{n}\rangle$ with $n_i = 0, 1$ and

$$[\hat{n}_i, \hat{n}_j] = 0. \tag{57}$$

2.3 The transformation

The JWT introduces new operators a_j by writing

$$c_j \equiv \left(\prod_{i=1}^{j-1} Z_i \right) a_j \qquad \& \qquad c_j^\dagger \equiv a_j^\dagger \left(\prod_{i=1}^{j-1} Z_i \right), \tag{58}$$

with $Z_j = (-1)^{n_j} = 1 - 2n_j$ and $n_j = c_j^\dagger c_j$. The ordering of the Z_i does not matter since $[Z_i, Z_j] = 0$ due to eq. (57). Inverting eq. (58) using $Z_i^2 = \mathbb{1}$ and $[Z_i, Z_j] = 0$ yields

$$a_j = \left(\prod_{i=1}^{j-1} Z_i\right) c_j \qquad \& \qquad a_j^\dagger = c_j^\dagger \left(\prod_{i=1}^{j-1} Z_i\right). \tag{59}$$

2.4 Algebra of the new operators

One can show that these new operators obey

$$[a_i^{(+)}, a_j^{(+)}] = 0 \qquad \forall\, i, j, \tag{60}$$

$$[a_i, a_j^\dagger] = 0 \qquad \forall\, i \neq j, \tag{61}$$

$$[a_i, a_i^\dagger] = 1 - 2n_i \qquad \text{with } n_i = c_i^\dagger c_i, \tag{62}$$

using that

$$\{Z_i, c_j^{(+)}\} = 0 \tag{63}$$

The corresponding proofs can be found in section 2.8.

2.5 The occupation-number basis in the new operators

The basis states of the fermionic Hilbert space can be written as

$$\begin{aligned}
|n_1, n_2, \ldots, n_L\rangle &\equiv \left(\prod_{i=1}^{L} (c_i^\dagger)^{n_i}\right) |0\rangle && \text{w. } n_i = 0, 1 \\
&= \left\{\prod_{i=1}^{L} \left[a_i^\dagger \left(\prod_{j=1}^{i-1} Z_j\right)\right]^{n_i}\right\} |0\rangle && (c_1^\dagger = a_1^\dagger) \\
&= (a_1^\dagger)^{n_1} (a_2^\dagger Z_1)^{n_2} (a_3^\dagger Z_2 Z_1)^{n_3} \cdots (a_L^\dagger Z_{L-1} \cdots Z_1)^{n_L} |0\rangle \\
&= \left(\prod_{i=1}^{L} (a_i^\dagger)^{n_i}\right) (Z_1)^{n_2} (Z_2 Z_1)^{n_3} \cdots (Z_{L-1} \cdots Z_1)^{n_L} |0\rangle \\
&= \left(\prod_{i=1}^{L} (a_i^\dagger)^{n_i}\right) |0\rangle && \text{w. } n_i = 0, 1,
\end{aligned} \tag{64}$$

using $[Z_i, Z_j] = 0\ \forall i, j$, $[Z_i, c_j] = 0$ for $i \neq j$ (hence $[Z_i, a_j^\dagger] = 0$ for $i \neq j$), and $Z_j |0\rangle = |0\rangle$. The ordering is not important anymore because $[a_i^\dagger, a_j^\dagger] = 0$.

2.6 Action on basis states

One cannot define the action of $a_j\,(a_j^\dagger)$ locally, since the JW string $Z_1\cdots Z_{j-1}$ is non-local, so one has to examine them on the full many-body basis. For $n_\ell = 0$,

$$\begin{aligned} a_\ell\,|n_1,\ldots,n_\ell=0,\ldots,n_L\rangle &= a_\ell\left(\prod_{i=1}^{L}(a_i^\dagger)^{n_i}\right)|0\rangle \\ &\overset{[a_\ell,a_j^\dagger]=0}{=} \left(\prod_{i=1}^{L}(a_i^\dagger)^{n_i}\right)a_\ell\,|0\rangle = 0, \end{aligned} \tag{65}$$

since $a_\ell\,|0\rangle = c_\ell\,|0\rangle = 0$. For $n_\ell = 1$,

$$\begin{aligned} a_\ell\,|n_1,\ldots,n_\ell=1,\ldots,n_L\rangle &= a_\ell\left(\prod_{\substack{i=1\\ i\neq\ell}}^{L}(a_i^\dagger)^{n_i}\right)a_\ell^\dagger\,|0\rangle \\ &\overset{[a_\ell,a_j^\dagger]=0}{=} \left(\prod_{\substack{i=1\\ i\neq\ell}}^{L}(a_i^\dagger)^{n_i}\right)a_\ell a_\ell^\dagger\,|0\rangle = \left(\prod_{\substack{i=1\\ i\neq\ell}}^{L}(a_i^\dagger)^{n_i}\right)c_\ell c_\ell^\dagger\,|0\rangle \\ &= \left(\prod_{\substack{i=1\\ i\neq\ell}}^{L}(a_i^\dagger)^{n_i}\right)|0\rangle = |n_1,\ldots,n_\ell=0,\ldots,n_L\rangle\,. \end{aligned} \tag{66}$$

Using $[a_\ell^\dagger, a_j^\dagger] = 0$ and $(a_\ell^\dagger)^2 = 0$, one quickly finds

$$a_\ell^\dagger\,|n_1,\ldots,n_\ell=0,\ldots,n_L\rangle = |n_1,\ldots,n_\ell=1,\ldots,n_L\rangle\,, \tag{67}$$

$$a_\ell^\dagger\,|n_1,\ldots,n_\ell=1,\ldots,n_L\rangle = 0. \tag{68}$$

Thus $a_\ell\,(a_\ell^\dagger)$ destroys (creates) a particle in the many-body state *without* giving rise to an extra sign structure (as $c_\ell\,(c_\ell^\dagger)$ would do).

2.7 Identification with spins

We can thus identify $n_i = 0, 1$ with $n_i = \downarrow, \uparrow$ in the many-body state, and the new operators can be associated with spins by identifying

$$a_j = S_j^-, \qquad a_j^\dagger = S_j^+ \qquad \& \qquad n_j - \tfrac{1}{2} = S_j^z = \tfrac{1}{2}\,\sigma_j^z, \tag{69}$$

since $[S_j^-, S_j^+] = -2S_j^z$ for spins (see the quick refresher on spin, section 9). For instance

$$a_3^\dagger\,|0\rangle = S_3^+\,|\downarrow\downarrow\downarrow\rangle = |\uparrow\downarrow\downarrow\rangle\,.$$

Matrix elements do not change under the JWT, since it is only a rewriting of operators and a renaming of basis states.

2.8 Proofs about the JWT

Claim: $\{Z_j, c_j^{(+)}\} = 0$**.**

Proof. On the single site j, with $Z_j |0\rangle_j = |0\rangle_j$, $Z_j |1\rangle_j = -|1\rangle_j$, $c_j |0\rangle_j = 0$ and $c_j |1\rangle_j = |0\rangle_j$,

$$Z_j c_j |0\rangle_j = Z_j \cdot 0 = 0 = c_j |0\rangle_j = c_j Z_j |0\rangle_j , \tag{70}$$

$$Z_j c_j |1\rangle_j = Z_j |0\rangle_j = |0\rangle_j = c_j |1\rangle_j , \tag{71}$$

$$c_j Z_j |1\rangle_j = -c_j |1\rangle_j = -|0\rangle_j . \tag{72}$$

Hence, on both basis states $\{Z_j, c_j\} = Z_j c_j + c_j Z_j = c_j - c_j = 0$, and since $Z_j^\dagger = Z_j$ it follows that $\{Z_j, c_j^\dagger\} = 0$ as well. (see Coleman, Chapter 4.1) □

Claim: $[a_i^{(+)}, a_j^{(+)}] = 0 \;\; \forall\, i, j$**.**

Proof. Assume w.l.o.g. that $i < j$. Using eq. (59),

$$a_j a_i = (Z_1 \cdots Z_{j-1} c_j)(Z_1 \cdots Z_{i-1} c_i)$$
$$\overset{\substack{[Z_j, c_\ell]=0,\, j\neq\ell \\ \{c_i, c_j\}=0}}{=} Z_1^2 \cdots Z_{i-1}^2 \, Z_i \cdots Z_{j-1} \, c_j c_i = -Z_i \cdots Z_{j-1} \, c_i c_j \qquad (= 0 \text{ if } i = j), \tag{73}$$

and likewise

$$a_i a_j = (Z_1 \cdots Z_{i-1} c_i)(Z_1 \cdots Z_{j-1} c_j)$$
$$\overset{[Z_i, c_\ell]=0,\, i\neq\ell}{=} Z_{i+1} \cdots Z_{j-1} \, c_i Z_i c_j \overset{\{Z_i, c_i\}=0}{=} -Z_i \cdots Z_{j-1} \, c_i c_j . \tag{74}$$

Comparing the two, $a_i a_j = a_j a_i$, i.e. $[a_i, a_j] = 0$. Analogously for $[a_i^\dagger, a_j^\dagger]$ and for $i > j$; the case $i = j$ is straightforward since $c_i^2 = 0$. □

Claim: $[a_i, a_j^\dagger] = 0 \;\; \forall\, i \neq j$**.**

Proof. We can show it w.l.o.g. for $i < j$. Using eq. (59) and eq. (63),

$$a_i^\dagger a_j = \big(c_i^\dagger Z_{i-1} \cdots Z_1\big)\big(Z_1 \cdots Z_{j-1} c_j\big) \overset{Z_i^2=1,\, i<j}{=} c_i^\dagger \, Z_i \cdots Z_{j-1} \, c_j$$
$$= Z_{i+1} \cdots Z_{j-1} \, c_i^\dagger Z_i c_j \overset{\{Z_i, c_i^\dagger\}=0}{=} -Z_i \cdots Z_{j-1} \, c_i^\dagger c_j = +Z_i \cdots Z_{j-1} \, c_j c_i^\dagger , \tag{75}$$

and

$$a_j a_i^\dagger = \big(Z_1 \cdots Z_{j-1} c_j\big)\big(c_i^\dagger Z_{i-1} \cdots Z_1\big)$$
$$= Z_i \cdots Z_{j-1} \, c_j c_i^\dagger \, Z_{i-1}^2 \cdots Z_1^2 \qquad ([Z_j, c_\ell^{(+)}] = 0,\; j \neq \ell)$$
$$= Z_i \cdots Z_{j-1} \, c_j c_i^\dagger . \tag{76}$$

Both reduce to the same expression, so $a_j a_i^\dagger - a_i^\dagger a_j = 0$ for $i < j$, i.e. $[a_j, a_i^\dagger] = 0$; the case $i > j$ follows straightforwardly. □

Claim: $[a_i, a_i^\dagger] = 1 - 2n_i$ **with** $n_i = c_i^\dagger c_i$**.**

Proof. Using eq. (59) and $Z_k^2 = \mathbb{1}$,

$$a_i^\dagger a_i = c_i^\dagger \, Z_{i-1} \cdots Z_1 Z_1 \cdots Z_{i-1} \, c_i = c_i^\dagger c_i = n_i, \tag{77}$$

$$a_i a_i^\dagger = Z_1 \cdots Z_{i-1} \, c_i c_i^\dagger \, Z_{i-1} \cdots Z_1 = c_i c_i^\dagger = 1 - c_i^\dagger c_i, \tag{78}$$

so that

$$[\hat{a}_i, \hat{a}_i^\dagger] = a_i a_i^\dagger - a_i^\dagger a_i = (1 - c_i^\dagger c_i) - c_i^\dagger c_i = 1 - 2\hat{n}_i. \tag{79}$$

□

2.9 References

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- *Introduction to Many-Body Physics*, Chapter 4.1 - Piers Coleman,
- *The Fermionic canonical commutation relations and the Jordan-Wigner transform* - M. Nielsen,

3 From the electron-phonon interaction to the BCS ground state

3.1 Overview

The road from deriving the Hamiltonian of the electron–phonon interaction to the BCS Hamiltonian, the physics it contains and finally its ground state is quite long and necessitates many calculations. Hence, this note only gives a concise summary and then refers to the literature for more information. I mostly follow *Solid State Theory 2* by Gerd Czycholl; equation numbers of the form (C 5.7) refer to that book. The Hamiltonian used as a starting point is derived by Czycholl in Sec. 7.1 of his first book and also by Bruus and Flensberg in Ch. 3 of *Many-Body Quantum Theory in Condensed Matter Physics*. If you are simply interested in the BCS ground state, you can directly skip to section 3.1.4. If you want the details, refer to section 3.2.

3.1.1 Attractive electron–electron interaction through the electron-phonon mechanism

Intuitive picture/motivation.

An electron e_1 moving through the lattice distorts it, i.e. it drags the positive ions towards itself and leaves behind a region of enhanced positive charge. A second electron e_2 then feels an attraction to e_1 through this phonon excitation caused by e_1. The result is an effective electron–electron interaction that is attractive.

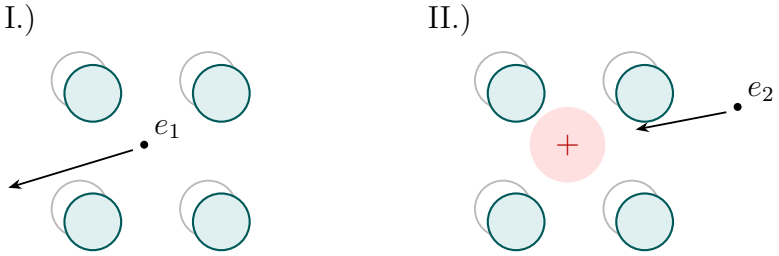


Figure 2: The lattice distortion caused by e_1 (I.) outlives its passage and attracts a second electron e_2 (II.).

Starting point.

The starting point for all our derivations is the Fröhlich Hamiltonian, derived in Czycholl Band 1, Sec. 7.1 (7.13)–(7.15) or Bruus and Flensberg, Sec. 3.8, (C 5.7):

$$H = \sum_{\boldsymbol{k},\sigma} \varepsilon_{\boldsymbol{k}}\, c^\dagger_{\boldsymbol{k},\sigma} c_{\boldsymbol{k},\sigma} + \sum_{\boldsymbol{q}} \hbar\omega_{\boldsymbol{q}}\, b^\dagger_{\boldsymbol{q}} b_{\boldsymbol{q}} + \sum_{\boldsymbol{k},\boldsymbol{q},\sigma} \Big[M(\boldsymbol{q})\, c^\dagger_{\boldsymbol{k}+\boldsymbol{q},\sigma} c_{\boldsymbol{k},\sigma} b_{\boldsymbol{q}} + M(-\boldsymbol{q})\, c^\dagger_{\boldsymbol{k}-\boldsymbol{q},\sigma} c_{\boldsymbol{k},\sigma} b^\dagger_{\boldsymbol{q}} \Big], \quad (80)$$

with $\omega_{-\boldsymbol{q}} = \omega_{\boldsymbol{q}}$ and $M(-\boldsymbol{q}) = M(\boldsymbol{q})$.

Canonical transformation.

Use a canonical transformation[3] $H \rightsquigarrow H_T = e^{-iS} H e^{iS}$ to find a purely electronic Hamiltonian that describes this interaction, i.e. to get rid of processes that change the number of phonons. At one point one simply restricts the Hamiltonian not to mix states with different phonon configurations.

The canonical transformation yields $H_T \cong H_0 + H_{\text{int}}$ with the interaction (C 5.25)

$$H_{\text{int}} = -\frac{1}{2} \sum_{\sigma,\sigma'} \sum_{\boldsymbol{k},\boldsymbol{k}',\boldsymbol{q}} V_{\boldsymbol{k},\boldsymbol{k}',\boldsymbol{q}}\; c^\dagger_{\boldsymbol{k}+\boldsymbol{q},\sigma}\, c^\dagger_{\boldsymbol{k}'-\boldsymbol{q},\sigma'}\, c_{\boldsymbol{k}',\sigma'}\, c_{\boldsymbol{k},\sigma}, \tag{81}$$

which is attractive if $|\varepsilon_{\boldsymbol{k}} - \varepsilon_{\boldsymbol{k}-\boldsymbol{q}}| \leq \hbar\omega_{\boldsymbol{q}} \leq \hbar\omega_D$ in $V_{\boldsymbol{k},\boldsymbol{k}',\boldsymbol{q}}$, where ω_D is the Debye frequency, and can be represented as in fig. 3.

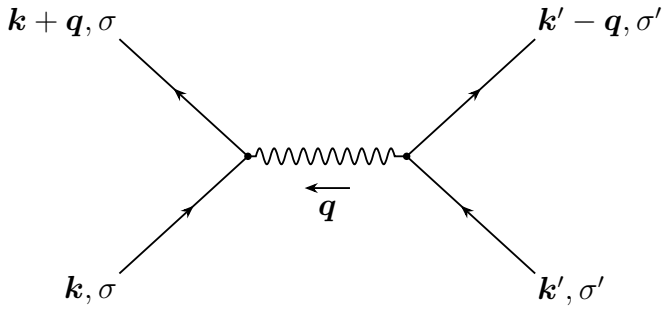


Figure 3: Vertex corresponding to eq. (81): two electrons exchange a phonon of momentum $\boldsymbol{q}$.

Since we know that the momentum transfer is $\boldsymbol{q}$, we know that $(\boldsymbol{k},\sigma) \to (\boldsymbol{k}'-\boldsymbol{q},\sigma')$ is not a valid fermion line.

3.1.2 Cooper pairs

In order to examine how this attractive interaction might change the many-body ground state, one looks at a highly simplified model:

- a system of free, non-interacting electrons, i.e. the Fermi sea (C 5.27)

$$|\phi_0\rangle = \prod_{\varepsilon_{\boldsymbol{k}} \leq \varepsilon_F,\,\sigma} c^\dagger_{\boldsymbol{k},\sigma} |0\rangle, \tag{82}$$

- the effective interaction is only present for two electrons that are placed right outside of the Fermi sea with opposite momenta and opposite spins (Cooper pair, singlet pairing);
- the interaction is $V = \text{const.}$ if $|\varepsilon_{\boldsymbol{k}+\boldsymbol{q}} - \varepsilon_{\boldsymbol{k}}| \leq \hbar\omega_D$ and zero otherwise.

[3]See section 4 for more general information on similarity transformations.

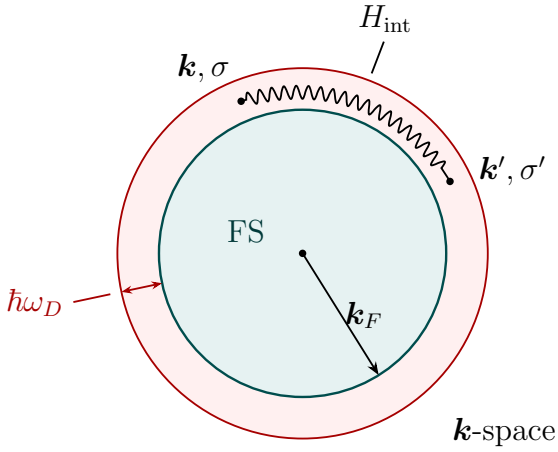


Figure 4: Only states in a shell of width $\hbar\omega_D$ around the Fermi surface take part in the effective interaction.

Here "right outside the Fermi sea" means that the energies are, cf. (C 5.27),

$$E_F \leq \varepsilon_{\boldsymbol{k}}, \;\; \varepsilon_{-\boldsymbol{k}} \leq E_F + \hbar\omega_D. \tag{83}$$

The interaction thus reads (C 5.28)

$$H_{\text{int}} = -\frac{V}{2} \sum_{\sigma,\sigma'} \sum_{\boldsymbol{k},\boldsymbol{q}} c^\dagger_{\boldsymbol{k}+\boldsymbol{q},\sigma}\, c^\dagger_{-\boldsymbol{k}-\boldsymbol{q},\sigma'}\, c_{-\boldsymbol{k},\sigma'}\, c_{\boldsymbol{k},\sigma}, \qquad \text{with } |\varepsilon_{\boldsymbol{k}+\boldsymbol{q}} - \varepsilon_{\boldsymbol{k}}| \leq \hbar\omega_D. \tag{84}$$

One solves this model by making the variational ansatz (C 5.26)

$$|\psi_{\sigma,\sigma'}\rangle = \sum_{\boldsymbol{k}} a_{\sigma,\sigma'}(\boldsymbol{k})\, c^\dagger_{\boldsymbol{k},\sigma}\, c^\dagger_{-\boldsymbol{k},\sigma'}\, |\phi_0\rangle \tag{85}$$

and minimizing the energy $E = \langle\psi_{\sigma,\sigma'}|H|\psi_{\sigma,\sigma'}\rangle$ with the constraint that $\langle\psi_{\sigma,\sigma'}|\psi_{\sigma,\sigma'}\rangle \stackrel{!}{=} 1$, which gets us an upper bound for the energy of the bound state. One finds that (C 5.35)

$$\Delta E = 2E_F - E_{\text{bs}} \cong -2\hbar\omega_{\boldsymbol{q}}\, e^{-2/(V\rho_0)}, \tag{86}$$

which shows that a bound state (Cooper pair) of the two electrons is energetically favoured compared to the free, non-interacting state. This holds for all electrons close to the Fermi edge and therefore implies an instability of the Fermi sea.

3.1.3 BCS theory

To solve the Hamiltonian and extract the relevant mechanism from it, it is simplified as follows: Only include interactions between electrons with opposite momenta and opposite spins, and neglect the $\boldsymbol{q}$-dependence. This yields (C 5.38)

$$H = \sum_{\boldsymbol{k},\sigma} \varepsilon_{\boldsymbol{k}}\, c^\dagger_{\boldsymbol{k}\sigma} c_{\boldsymbol{k}\sigma} - V \sum_{\boldsymbol{k},\boldsymbol{k}'} c^\dagger_{\boldsymbol{k}'\uparrow} c^\dagger_{-\boldsymbol{k}'\downarrow} c_{-\boldsymbol{k}\downarrow} c_{\boldsymbol{k}\uparrow}, \tag{87}$$

with again $|\varepsilon_{\boldsymbol{k}} - \varepsilon_{\boldsymbol{k}'}| \leq \hbar\omega_D$ for the interaction term. Finally, apply mean-field theory to the pair creation/destruction operators, (C 5.41) and (C 5.42):

$$H_{\mathrm{MF}} = \sum_{\boldsymbol{k}} \begin{pmatrix} c^\dagger_{\boldsymbol{k}\uparrow} & c_{-\boldsymbol{k}\downarrow} \end{pmatrix} \begin{pmatrix} \varepsilon_{\boldsymbol{k}} & -\Delta \\ -\Delta^* & -\varepsilon_{\boldsymbol{k}} \end{pmatrix} \begin{pmatrix} c_{\boldsymbol{k}\uparrow} \\ c^\dagger_{-\boldsymbol{k}\downarrow} \end{pmatrix} \pm \text{ const.}, \tag{88}$$

$$\Delta = V \sum_{\boldsymbol{k}} \langle c_{-\boldsymbol{k}\downarrow} c_{\boldsymbol{k}\uparrow} \rangle \qquad (\text{possibly } \varepsilon_{\boldsymbol{k}} \rightsquigarrow \varepsilon_{\boldsymbol{k}} - \mu). \tag{89}$$

Bogoliubov transformation.

Introduce a unitary transformation that diagonalizes

$$B = \begin{pmatrix} \varepsilon_{\boldsymbol{k}} & -\Delta \\ -\Delta^* & -\varepsilon_{\boldsymbol{k}} \end{pmatrix}$$

and thus leads to an H_{MF} that only includes terms of the form $\alpha^\dagger_{\boldsymbol{k}}\alpha_{\boldsymbol{k}}$ and $\beta^\dagger_{\boldsymbol{k}}\beta_{\boldsymbol{k}}$:

$$H_{\mathrm{MF}} = \sum_{\boldsymbol{k}} \underbrace{\begin{pmatrix} c^\dagger_{\boldsymbol{k}\uparrow} & c_{-\boldsymbol{k}\downarrow} \end{pmatrix} U_{\boldsymbol{k}}}_{\text{new operators}} \underbrace{U^\dagger_{\boldsymbol{k}} \begin{pmatrix} \varepsilon_{\boldsymbol{k}} & -\Delta \\ -\Delta^* & -\varepsilon_{\boldsymbol{k}} \end{pmatrix} U_{\boldsymbol{k}}}_{\text{diagonal matrix}} \underbrace{U^\dagger_{\boldsymbol{k}} \begin{pmatrix} c_{\boldsymbol{k}\uparrow} \\ c^\dagger_{-\boldsymbol{k}\downarrow} \end{pmatrix}}_{(\text{new operators})^\dagger} . \tag{90}$$

These new operators will be fermionic, since U is unitary, so that $\langle c_{-\boldsymbol{k}\downarrow} c_{\boldsymbol{k}\uparrow} \rangle \rightsquigarrow f(\alpha_{\boldsymbol{k}}, \beta_{\boldsymbol{k}}, \alpha^\dagger_{\boldsymbol{k}}, \beta^\dagger_{\boldsymbol{k}})$ and one can then use $\langle \alpha^\dagger_{\boldsymbol{k}}\alpha_{\boldsymbol{k}} \rangle = f(E_F)$ and $\langle \beta_{\boldsymbol{k}}\alpha_{\boldsymbol{k}} \rangle = 0$ etc. The elements of U can be determined by diagonalizing B and requiring that the eigenvectors are normalized to one.

Result.

This yields

$$H_{\mathrm{eff}} = \sum_{\boldsymbol{k}} E_{\boldsymbol{k}} \big(\alpha^\dagger_{\boldsymbol{k}}\alpha_{\boldsymbol{k}} + \beta^\dagger_{\boldsymbol{k}}\beta_{\boldsymbol{k}} \big) + \text{const.}, \qquad (\text{C } 5.46) \tag{91}$$

$$E_{\boldsymbol{k}} = \big(\varepsilon^2_{\boldsymbol{k}} + |\Delta|^2 \big)^{1/2} \ \geq \ 0, \qquad (\text{C } 5.47) \tag{92}$$

and for Δ one arrives at the BCS self-consistency equation (C 5.50)

$$\Delta = \frac{V\Delta}{2} \sum_{\boldsymbol{k}} \frac{1}{E_{\boldsymbol{k}}} \tanh\left(\frac{\beta E_{\boldsymbol{k}}}{2} \right). \tag{93}$$

Obviously $\Delta = 0$ is a solution. For $\Delta \neq 0$ one obtains (C 5.51)

$$1 = \frac{V}{2} \sum_{\boldsymbol{k}, |\varepsilon_{\boldsymbol{k}}| < \hbar\omega_D} \big(\varepsilon^2_{\boldsymbol{k}} + |\Delta|^2 \big)^{-1/2} \tanh\left(\frac{\beta \big(\varepsilon^2_{\boldsymbol{k}} + |\Delta|^2 \big)^{1/2}}{2} \right). \tag{94}$$

Δ as an order parameter.

Actually, Δ is an order parameter for the superconducting phase, because of the following:

- For a normal metal $\langle c_{\boldsymbol{k}\uparrow} c_{-\boldsymbol{k}\downarrow}\rangle = 0 \;\forall\, \boldsymbol{k}$ and thus $\Delta = V \sum_{\boldsymbol{k}} \langle c_{-\boldsymbol{k}\downarrow} c_{\boldsymbol{k}\uparrow}\rangle = 0$.
- But if $\Delta \neq 0$, it means that we are not in a metallic phase and that there are Cooper pairs present.

3.1.4 The BCS ground state

We have $H = \sum_{\boldsymbol{k}} E_{\boldsymbol{k}} (\alpha^\dagger_{\boldsymbol{k}} \alpha_{\boldsymbol{k}} + \beta^\dagger_{\boldsymbol{k}} \beta_{\boldsymbol{k}})$ with $E_{\boldsymbol{k}} \geq 0$, cf. eq. (91), where, following (C 5.43),

$$\alpha_{\boldsymbol{k}} = u_{\boldsymbol{k}} c_{\boldsymbol{k}\uparrow} - v_{\boldsymbol{k}} c^\dagger_{-\boldsymbol{k}\downarrow} \qquad \text{and} \qquad \beta_{\boldsymbol{k}} = u_{\boldsymbol{k}} c_{-\boldsymbol{k}\downarrow} + v_{\boldsymbol{k}} c^\dagger_{\boldsymbol{k}\uparrow}. \tag{95}$$

Since $\alpha^{(\dagger)}_{\boldsymbol{k}}$ and $\beta^{(\dagger)}_{\boldsymbol{k}}$ are fermionic, the eigenvalues of $\alpha^\dagger_{\boldsymbol{k}} \alpha_{\boldsymbol{k}}$ (and $\beta^\dagger_{\boldsymbol{k}} \beta_{\boldsymbol{k}}$) are $0, 1$. Hence the ground state of H does not contain any α or β quasiparticles, i.e.

$$\alpha_{\boldsymbol{k}} |\mathrm{BCS}\rangle = \beta_{\boldsymbol{k}} |\mathrm{BCS}\rangle = 0. \tag{96}$$

This is satisfied by

$$|\mathrm{BCS}\rangle = \prod_{\boldsymbol{k}} \left(u_{\boldsymbol{k}} + v_{\boldsymbol{k}} c^\dagger_{\boldsymbol{k}\uparrow} c^\dagger_{-\boldsymbol{k}\downarrow}\right)|0\rangle = \bigotimes_{\boldsymbol{k}} \left(u_{\boldsymbol{k}} + v_{\boldsymbol{k}} c^\dagger_{\boldsymbol{k}\uparrow} c^\dagger_{-\boldsymbol{k}\downarrow}\right)|0\rangle_{\boldsymbol{k}}. \tag{97}$$

3.2 Details

3.2.1 Preface

These are supplementary notes to *Solid State Theory, Volume 2* by Gerd Czycholl. Since the derivations there are quite lengthy, only those steps are presented here that clarify what is done in the corresponding chapters. Numbers in parentheses such as (5.12) refer to equations of that book.

3.2.2 Attractive electron-electron interaction through the electron-phonon mechanism (Czycholl 5.2)

Two small observations.

- (5.12): use that S is linear in H_1 to discard $[[H_1, S], S]$.
- (5.15): use that $\langle m|H_1|m\rangle = 0$, since H_1 changes the number of phonons and thus $H_1|m\rangle \perp |m\rangle$.

The transformed Hamiltonian.

Starting from (5.13) and inserting resolutions of the identity,

$$H_T = H_0 + \frac{i}{2}\big(H_1 S - S H_1\big) \tag{98}$$

$$= H_0 + \frac{i}{2}\sum_{\tilde{n},n,m}\Big(|\tilde{n}\rangle\langle\tilde{n}|H_1|m\rangle\langle m|S|n\rangle\langle n| - |\tilde{n}\rangle\langle\tilde{n}|S|m\rangle\langle m|H_1|n\rangle\langle n|\Big) \tag{99}$$

$$\overset{(5.15)}{=} H_0 - \frac{1}{2}\sum_{\tilde{n},n,m}\Big(|\tilde{n}\rangle\langle\tilde{n}|H_1|m\rangle\langle m|H_1|n\rangle\langle n|\frac{1}{E_m - E_n} - |\tilde{n}\rangle\langle\tilde{n}|H_1|m\rangle\langle m|H_1|n\rangle\langle n|\frac{1}{E_{\tilde{n}} - E_m}\Big), \tag{100}$$

where the matrix elements $\langle\tilde{n}|H_1|m\rangle$ and $\langle m|H_1|n\rangle$ are scalars and can be pulled out of the operator structure, which yields (5.16).

Restriction to a fixed phonon configuration.

Since we want to derive a Hamiltonian for the electron-electron interaction only, we restrict $|\tilde{n}\rangle$ and $|n\rangle$ in (5.16) to states with the same phonon configuration, such that H_T does not mix states with different phonon configurations.

Let $|n\rangle$ be some many-particle state and assume that it contains one electron in the state $|k,\sigma\rangle$ and one in $|k',\sigma'\rangle$. The two elementary processes are

$$H_{1,1}|k,\sigma\rangle = \sqrt{n_q}\,|k+q,\sigma,n_q-1\rangle, \qquad \text{by destroying a phonon,} \tag{$*_1$}$$

$$H_{1,2}|k,\sigma\rangle = \sqrt{n_q+1}\,|k-q,\sigma,n_q+1\rangle, \qquad \text{by creating a phonon.} \tag{$*_2$}$$

Since $|\tilde{n}\rangle$ and $|n\rangle$ are restricted to states with the same phonon number, the created (destroyed) phonon with momentum q has to be destroyed (created) again:

$$H_{1,1}\sqrt{n_q}\,|k+q,\sigma,n_q-1\rangle = n_q\,|k+q,\sigma;\,k'-q,\sigma';\,n_q\rangle, \tag{101}$$

$$H_{1,1}\sqrt{n_q+1}\,|k-q,\sigma,n_q+1\rangle = (n_q+1)\,|k-q,\sigma;\,k'+q,\sigma';\,n_q\rangle. \tag{102}$$

All other terms vanish, since their phonon configurations differ.

Energy denominators.

Therefore we have

$$(*_1): \quad E_m - E_n = \varepsilon_{k+q} - \varepsilon_k - \hbar\omega_q, \qquad E_{\tilde{n}} - E_m = \varepsilon_{k'-q} - \varepsilon_{k'} + \hbar\omega_q, \qquad (5.19) \tag{103}$$

$$(*_2): \quad E_m - E_n = \varepsilon_{k-q} - \varepsilon_k + \hbar\omega_q, \qquad E_{\tilde{n}} - E_m = \varepsilon_{k'+q} - \varepsilon_{k'} - \hbar\omega_q. \qquad (5.20) \tag{104}$$

Inserting this into (5.16) gives

$$
\begin{aligned}
H_T = -\frac{1}{2}\sum_{n,\tilde{n}} |\tilde{n}\rangle\langle n| \sum_m \sum_{\substack{k,k',q \\ \sigma,\sigma'}} \Big(& \langle \tilde{n}|M(-q)c^\dagger_{k'-q,\sigma'}c_{k',\sigma'}b^\dagger_q|m\rangle\langle m|M(q)c^\dagger_{k+q,\sigma}c_{k,\sigma}b_q|n\rangle \\
& \times \Big(\frac{1}{\varepsilon_{k+q}-\hbar\omega_q-\varepsilon_k} - \frac{1}{\varepsilon_{k'-q}-\varepsilon_{k'}+\hbar\omega_q}\Big) \\
& + \langle \tilde{n}|M(q)c^\dagger_{k'+q,\sigma'}c_{k',\sigma'}b_q|m\rangle\langle m|M(-q)c^\dagger_{k-q,\sigma}c_{k,\sigma}b^\dagger_q|n\rangle \\
& \times \Big(\frac{1}{\varepsilon_{k-q}-\varepsilon_k+\hbar\omega_q} - \frac{1}{\varepsilon_{k'+q}-\varepsilon_{k'}-\hbar\omega_q}\Big)\Big),
\end{aligned}
\tag{105}
$$

where the first summand is the $(*_1)$-process and the second one the $(*_2)$-process.

Using $M^*(q) = M(-q)$ and that all the above terms are independent of n, m, and $\tilde{n}$ yields (5.21). Using furthermore that

$$
b^\dagger_q b_q|\dots n_q \dots\rangle = n_q|\dots n_q\dots\rangle, \qquad b_q b^\dagger_q|\dots n_q\dots\rangle = (n_q+1)|\dots n_q\dots\rangle \tag{106}
$$

results in (5.21), i.e. $\hat{H}_T(b_q, b^\dagger_q, \dots) \rightsquigarrow \hat{H}_T(n_q, \dots)$ in the phonon number basis.

Combining the two processes.

We have

$$
\frac{1}{\varepsilon_{k+q}-\hbar\omega_q-\varepsilon_k} = \frac{(\varepsilon_{k+q}-\varepsilon_k)+\hbar\omega_q}{(\varepsilon_{k+q}-\varepsilon_k)^2-\hbar^2\omega_q^2} \tag{107}
$$

and, relabelling $k \leftrightarrow k'$ and $q \to -q$,

$$
\sum_{k,k',q} \frac{1}{\varepsilon_{k'-q}-\varepsilon_{k'}+\hbar\omega_q}\, c^\dagger_{k+q}c_k c^\dagger_{k'-q}c_{k'} = \sum_{k,k',q} \frac{(\varepsilon_{k+q}-\varepsilon_k)-\hbar\omega_q}{(\varepsilon_{k+q}-\varepsilon_k)^2-\hbar^2\omega_q^2}\, c^\dagger_{k'-q}c_{k'}c^\dagger_{k+q}c_k, \tag{108}
$$

where

$$
\begin{aligned}
\sum_{k,k',q} c^\dagger_{k'-q}c_{k'}c^\dagger_{k+q}c_k &= \sum_{k,k',q} c^\dagger_{k'-q}\big(\delta_{k',k+q} - c^\dagger_{k+q}c_{k'}\big)c_k && (109)\\
&= \sum_{k,q} c^\dagger_k c_k - \sum_{k,k',q} c^\dagger_{k+q}c^\dagger_{k'-q}c_k c_{k'} && (110)\\
&= \sum_{k,q} n_k - \sum_{k,k',q} c^\dagger_{k+q}\big(\delta_{k,k'-q} - c_k c^\dagger_{k'-q}\big)c_{k'} && (111)\\
&= \underbrace{\sum_{k,q} n_k - \sum_{k,q} c^\dagger_{k'}c_{k'}}_{=0} + \sum_{k,k',q} c^\dagger_{k+q}c_k c^\dagger_{k'-q}c_{k'} && (112)\\
&= \sum_{k,k',q} c^\dagger_{k+q}c_k c^\dagger_{k'-q}c_{k'}. && (113)
\end{aligned}
$$

We know (Bruus & Flensberg, Sec. 3.8) that $\omega_{-q} = \omega_q$ and $M(-q) = M(q)$. Furthermore, since we are only interested in the electronic process, we can assume that there is no net phononic motion within the system and thus $n_q = n_{-q}$. Hence the first term of (5.21) equals the first term of (5.22), analogously for the second term, and similarly (5.22) $\rightsquigarrow$ (5.23).

3.2.3 Cooper pairs (Czycholl 5.3)

Setup.

Simplify: look at the Fermi sea plus two electrons just above the Fermi energy, and only let them interact via H_T. The variational ansatz (5.26), for vanishing total pair momentum $K = k + k' \stackrel{!}{=} 0$, is

$$|\psi_{\sigma,\sigma'}\rangle = \sum_k a_{\sigma,\sigma'}(k)\, c^\dagger_{k,\sigma} c^\dagger_{-k,\sigma'} |\phi_0\rangle, \tag{114}$$

with the sum over k such that $E_F \le \varepsilon_k, \varepsilon_{k'} \le E_F + \hbar\omega_D$, cf. (5.27). The simplified interaction acts only outside the Fermi sea $(*)$, otherwise $V = 0$:

$$H_T = -\frac{V}{2} \sum_{\sigma,\sigma'} \sum_{k,q} c^\dagger_{k+q,\sigma} c^\dagger_{-k-q,\sigma'} c_{-k,\sigma'} c_{k,\sigma}. \qquad (5.28) \tag{115}$$

Action on the ansatz.

$$H_T|\psi_{\sigma_1\sigma_2}\rangle = -\frac{V}{2} \sum_{\sigma,\sigma'} \sum_{k,q,k'} c^\dagger_{k+q,\sigma} c^\dagger_{-k-q,\sigma'} c_{-k,\sigma'} c_{k,\sigma}\, a_{\sigma_1\sigma_2}(k')\, c^\dagger_{k',\sigma_1} c^\dagger_{-k',\sigma_2} |\phi_0\rangle \tag{116}$$

$$= -\frac{V}{2} \sum_{\sigma,\sigma'} \sum_{k,q,k'} a_{\sigma_1\sigma_2}(k')\, c^\dagger_{k+q,\sigma} c^\dagger_{-k-q,\sigma'} c_{-k,\sigma'} c_{k,\sigma}\, |k', \sigma_1; -k', \sigma_2\rangle \tag{117}$$

$$= -\frac{V}{2} \sum_{\sigma,\sigma'} \sum_{k,q,k'} a_{\sigma_1\sigma_2}(k') \Big(\delta_{k,k'}\delta_{\sigma,\sigma_1}\delta_{\sigma',\sigma_2} |k+q, \sigma_1; -k-q, \sigma_2\rangle + \delta_{k,-k'}\delta_{\sigma,\sigma_2}\delta_{\sigma',\sigma_1} |-k-q, \sigma_1; k+q, \sigma_2\rangle \Big) \tag{118}$$

$$= -\frac{V}{2} \sum_{k,q} \Big(a_{\sigma_1\sigma_2}(k)\, |k+q, \sigma_1; -k-q, \sigma_2\rangle + a_{\sigma_1\sigma_2}(-k)\, |-k-q, \sigma_1; k+q, \sigma_2\rangle \Big), \tag{119}$$

where the projection onto $|k', \sigma_1; -k', \sigma_2\rangle$ is allowed because, by $(*)$, only states outside the Fermi sea matter for the interaction. Taking the expectation value,

$$\langle\psi_{\sigma_1\sigma_2}|H_T|\psi_{\sigma_1\sigma_2}\rangle = -\frac{V}{2} \sum_{k,q,k'} a^*_{\sigma_1\sigma_2}(k') \Big\{ a_{\sigma_1\sigma_2}(k)\, \langle k', \sigma_1; -k', \sigma_2 | k+q, \sigma_1; -k-q, \sigma_2\rangle + a_{\sigma_1\sigma_2}(-k)\, \langle k', \sigma_1; -k', \sigma_2 | -k-q, \sigma_1; k+q, \sigma_2\rangle \Big\} \tag{120}$$

$$= -\frac{V}{2} \sum_{k,q,k'} a^*_{\sigma_1\sigma_2}(k') \Big\{ a_{\sigma_1\sigma_2}(k)\, \delta_{k',k+q} + a_{\sigma_1\sigma_2}(-k)\, \delta_{k',-k-q} \Big\} \tag{121}$$

$$= -\frac{V}{2} \sum_{k,q} \Big(a^*_{\sigma_1\sigma_2}(k+q) a_{\sigma_1\sigma_2}(k) + a^*_{\sigma_1\sigma_2}(-k-q) a_{\sigma_1\sigma_2}(-k) \Big) \tag{122}$$

$$= -V \sum_{k,q} a^*_{\sigma_1\sigma_2}(k+q) a_{\sigma_1\sigma_2}(k), \tag{123}$$

where in the last step $k \to -k$, $q \to -q$ was substituted in the second term.

The kinetic part.

Furthermore, by (5.17),

$$H_0 = \sum_{k,\sigma} \varepsilon_k n_{k,\sigma} + \sum_q \hbar\omega_q b_q^\dagger b_q. \tag{124}$$

Since the ansatz $|\psi_{\sigma\sigma'}\rangle$ is purely electronic, $\sum_q \hbar\omega_q b_q^\dagger b_q$ will not contribute. Splitting the electronic sum into states inside and outside the Fermi sea,

$$\langle\psi_{\sigma,\sigma'}|H_0|\psi_{\sigma,\sigma'}\rangle = \langle\psi_{\sigma\sigma'}|\Big(\sum_{k,\tilde\sigma} \varepsilon_k n_{k,\tilde\sigma}\Big)\Big(\sum_{k'} a_{\sigma,\sigma'}(k') c^\dagger_{k',\sigma} c^\dagger_{-k',\sigma'}|\phi_0\rangle\Big) \tag{125}$$

$$= \langle\psi_{\sigma\sigma'}|\Big(\sum_{\substack{k\le k_F\\ \tilde\sigma}} \varepsilon_k \hat n_{k,\tilde\sigma} + \sum_{\substack{k> k_F\\ \tilde\sigma}} \varepsilon_k \hat n_{k,\tilde\sigma}\Big)\Big(\sum_{k'} a_{\sigma,\sigma'}(k') c^\dagger_{k',\sigma} c^\dagger_{-k',\sigma'}|\phi_0\rangle\Big) \tag{126}$$

$$= \langle\psi_{\sigma\sigma'}| \sum_{\substack{k\le k_F\\ \tilde\sigma}} \varepsilon_k n_{k,\tilde\sigma}|\psi_{\sigma\sigma'}\rangle + \sum_{k,k'} a_{\sigma,\sigma'}(k')\big(\varepsilon_{k'}\delta_{k,k'}\delta_{\sigma,\tilde\sigma} + \varepsilon_{k'}\delta_{k,-k'}\delta_{\sigma',\tilde\sigma}\big) c^\dagger_{k',\sigma} c^\dagger_{-k',\sigma'}|\phi_0\rangle \tag{127}$$

$$= \Big(\langle\phi_0| \sum_k c_{-k,\sigma'} c_{k,\sigma} a^*_{\sigma,\sigma'}(k)\Big)\Big(\sum_{\substack{k\le k_F\\ \tilde\sigma}} \varepsilon_k n_{k,\tilde\sigma}|\psi_{\sigma\sigma'}\rangle + \sum_{k'} a_{\sigma,\sigma'}(k')(\varepsilon_{k'} + \varepsilon_{-k'}) c^\dagger_{k',\sigma} c^\dagger_{-k',\sigma'}|\phi_0\rangle\Big) \tag{128}$$

$$= \sum_k |a_{\sigma,\sigma'}(k)|^2\Big((\varepsilon_k + \varepsilon_{-k}) + \sum_{\substack{k\le k_F\\ \tilde\sigma}} \varepsilon_k n_{k,\tilde\sigma}\Big) \tag{129}$$

$$= \sum_k |a_{\sigma,\sigma'}(k)|^2(\varepsilon_k + \varepsilon_{-k}) + c\sum_k |a_{\sigma\sigma'}(k)|^2, \tag{130}$$

where c collects the (constant) energy of the filled Fermi sea. From $\langle\psi_{\sigma\sigma'}|\psi_{\sigma\sigma'}\rangle \overset{!}{=} 1$ it follows that $\sum_k |a_{\sigma\sigma'}(k)|^2 \overset{!}{=} 1$, cf. (5.29), and therefore

$$H_0 = \sum_k |a_{\sigma,\sigma'}(k)|^2(\varepsilon_k + \varepsilon_{-k}), \qquad \Longrightarrow \qquad H = H_0 + H_T = (5.29). \tag{131}$$

The step $\varepsilon_k = \varepsilon_{-k}$ needed for (5.30) is true due to Kramers' theorem; see e.g. Sec. 5.3 of Czycholl I.

When does the gap equation degenerate?

For (5.32) it was assumed that $C \equiv \sum_k a_{\sigma\sigma'}(k) \neq 0$. Let us examine when C is actually zero. We have

$$a_{\sigma\sigma'}(k) = \frac{V\sum_{k'} a_{\sigma\sigma'}(k')}{2\varepsilon_k - \lambda}, \qquad (5.31) \tag{132}$$

where $\varepsilon_k = \varepsilon_{-k}$ by Kramers' theorem, and hence

$$a_{\sigma\sigma'}(-k) = a_{\sigma\sigma'}(k). \tag{$*$}$$

Moreover, for equal spins,

$$|\psi_{\sigma\sigma}\rangle = \sum_k a_{\sigma,\sigma}(k) c^\dagger_{k,\sigma} c^\dagger_{-k,\sigma} |\phi_0\rangle \tag{133}$$

$$= \sum_k a_{\sigma,\sigma}(k) \big(\delta_{k,-k} - c^\dagger_{-k,\sigma} c^\dagger_{k,\sigma}\big) |\phi_0\rangle \tag{134}$$

$$= -\sum_k a_{\sigma,\sigma}(k) c^\dagger_{-k,\sigma} c^\dagger_{k,\sigma} |\phi_0\rangle \tag{135}$$

$$= -\sum_k a_{\sigma,\sigma}(-k) c^\dagger_{k,\sigma} c^\dagger_{-k,\sigma} |\phi_0\rangle \tag{136}$$

$$= -\sum_k a_{\sigma,\sigma}(k) c^\dagger_{k,\sigma} c^\dagger_{-k,\sigma} |\phi_0\rangle = -|\psi_{\sigma\sigma}\rangle. \tag{137}$$

Here $\delta_{k,-k}$ drops out because $\sum_k$ does not include momenta inside the Fermi sea and thus $k \neq 0$; in the fourth line $k \to -k$ was substituted, and in the fifth line $(*)$ was used. Consequently

$$\langle\psi_{\sigma\sigma}|\psi_{\sigma\sigma}\rangle = -\langle\psi_{\sigma\sigma}|\psi_{\sigma\sigma}\rangle \implies \langle\psi_{\sigma\sigma}|\psi_{\sigma\sigma}\rangle = 0 \implies \sum_k |a_{\sigma\sigma}(k)|^2 = 0 \implies a_{\sigma\sigma}(k) = 0, \tag{138}$$

so $\sum_k a_{\sigma\sigma}(k) = 0$. Thus $\sigma = \sigma'$ yields $C = 0$, cf. (5.35), and Cooper pairs consist of electrons with opposite spins.

Note that if $\psi(r_1 - r_2)$ is antisymmetric, then $\psi(0) = 0$ and $C = 0$ as well, so the derivation of (5.35) is different in that case.

Position representation.

Project the state $|\psi_{\sigma,\sigma'}\rangle = \sum_k a_{\sigma,\sigma'}(k) c^\dagger_{k,\sigma} c^\dagger_{-k,\sigma'} |\phi_0\rangle$ onto the position basis. Since H acts trivially on the spin space ($\sigma \to \sigma$, $\sigma' \to \sigma'$), one can factorize the wave function:

$$\langle \boldsymbol{r}_1, \boldsymbol{r}_2 | \psi_{\sigma,\sigma'}\rangle = \psi_{\sigma,\sigma'}(\boldsymbol{r}_1, \boldsymbol{r}_2) = \psi(\boldsymbol{r}_1, \boldsymbol{r}_2)\, \chi(\sigma, \sigma'), \qquad (5.37) \tag{139}$$

with

$$\psi(\boldsymbol{r}_1, \boldsymbol{r}_2) = \frac{1}{V} \sum_k a(k)\, e^{i\boldsymbol{k}\cdot\boldsymbol{r}_1} e^{-i\boldsymbol{k}\cdot\boldsymbol{r}_2} = \psi(\boldsymbol{r}_1 - \boldsymbol{r}_2), \qquad (5.36) \tag{140}$$

using $\langle \boldsymbol{r}|\boldsymbol{k}\rangle = \frac{1}{\sqrt{V}} e^{i\boldsymbol{k}\cdot\boldsymbol{r}}$.

3.2.4 BCS theory (Czycholl 5.4)

Mean-field recap.

Write $\hat{n}_\alpha = \langle n_\alpha\rangle + \delta\hat{n}_\alpha$ and assume that $\delta\hat{n}_\alpha$ is small, such that second-order fluctuations can be dropped:

$$\hat{n}_\alpha \hat{n}_\beta = \big(\langle\hat{n}_\alpha\rangle + \delta\hat{n}_\alpha\big)\big(\langle\hat{n}_\beta\rangle + \delta\hat{n}_\beta\big) \tag{141}$$

$$\overset{\delta\hat{n}_\alpha\delta\hat{n}_\beta \approx 0}{=} \langle\hat{n}_\alpha\rangle\langle\hat{n}_\beta\rangle + \langle\hat{n}_\alpha\rangle\big(\hat{n}_\beta - \langle\hat{n}_\beta\rangle\big) + \langle\hat{n}_\beta\rangle\big(\hat{n}_\alpha - \langle\hat{n}_\alpha\rangle\big) \tag{142}$$

$$= \langle\hat{n}_\alpha\rangle\hat{n}_\beta + \langle\hat{n}_\beta\rangle\hat{n}_\alpha - \langle\hat{n}_\alpha\rangle\langle\hat{n}_\beta\rangle. \tag{143}$$

Now apply this to $c^{(\dagger)}_{-k\downarrow}c^{(\dagger)}_{k\uparrow}$ instead of $\hat{n}_\alpha$ to derive a mean-field version of the BCS Hamiltonian, which gives (5.40).

Nambu form of the effective Hamiltonian.

Rewrite H_{eff} as (following Becca & Sorella, and the Superconductivity IMPRS UFAST 2017 course of Michael Sentef)

$$\tilde{H}_{\text{eff}} = \sum_k \begin{pmatrix} c^\dagger_{k\uparrow} & c_{-k\downarrow} \end{pmatrix} \begin{pmatrix} \varepsilon_k & -\Delta \\ -\Delta^* & -\varepsilon_k \end{pmatrix} \begin{pmatrix} c_{k\uparrow} \\ c^\dagger_{-k\downarrow} \end{pmatrix} \tag{144}$$

$$= \sum_k \begin{pmatrix} c^\dagger_{k\uparrow} & c_{-k\downarrow} \end{pmatrix} \begin{pmatrix} \varepsilon_k c_{k\uparrow} - \Delta c^\dagger_{-k\downarrow} \\ -\Delta^* c_{k\uparrow} - \varepsilon_k c^\dagger_{-k\downarrow} \end{pmatrix} \tag{145}$$

$$= \sum_k \varepsilon_k c^\dagger_{k\uparrow}c_{k\uparrow} - \Delta c^\dagger_{k\uparrow}c^\dagger_{-k\downarrow} - \Delta^* c_{-k\downarrow}c_{k\uparrow} - \varepsilon_k \underbrace{c_{-k\downarrow}c^\dagger_{-k\downarrow}}_{=1-c^\dagger_{-k\downarrow}c_{-k\downarrow}} \tag{146}$$

$$= \sum_{k,\sigma} \varepsilon_k c^\dagger_{k\sigma}c_{k\sigma} - \Delta^* \sum_k c_{-k\downarrow}c_{k\uparrow} - \Delta \sum_k c^\dagger_{k\uparrow}c^\dagger_{-k\downarrow} - \sum_k \varepsilon_k \tag{147}$$

$$= H_{\text{eff}} - \underbrace{\frac{|\Delta|^2}{V} - \sum_k \varepsilon_k}_{\text{const.}}, \qquad (5.41) \tag{148}$$

Set $M \equiv \begin{pmatrix} \varepsilon_k & -\Delta \\ -\Delta^* & -\varepsilon_k \end{pmatrix}$ and diagonalize it, such that H_{eff} becomes diagonal — this is a unitary transformation.

Diagonalizing the 2×2 matrix M.

Eigenvalues:

$$\det \begin{pmatrix} \varepsilon_k - \lambda & -\Delta \\ -\Delta^* & -\varepsilon_k - \lambda \end{pmatrix} = -(\varepsilon_k - \lambda)(\varepsilon_k + \lambda) - |\Delta|^2 \tag{149}$$

$$= -\big(\lambda^2 - \varepsilon_k^2 - |\Delta|^2\big) \stackrel{!}{=} 0 \tag{150}$$

$$\implies \quad \lambda_{1,2} = \pm\sqrt{\varepsilon_k^2 + |\Delta|^2} \equiv E_k^\pm \in \mathbb{R}. \tag{$*_1$}$$

Eigenvectors: from

$$\begin{pmatrix} \varepsilon_k - E_k^\pm & -\Delta \\ -\Delta^* & -\varepsilon_k - E_k^\pm \end{pmatrix} \begin{pmatrix} x_\pm \\ y_\pm \end{pmatrix} \stackrel{!}{=} 0 \tag{151}$$

the first row (I) gives

$$(\varepsilon_k - E_k^\pm)x_\pm - \Delta y_\pm = 0 \quad \iff \quad y_\pm = \frac{\varepsilon_k - E_k^\pm}{\Delta}\, x_\pm, \tag{I$'$}$$

while the second row (II) gives, using (I$'$) and $(E_k^\pm)^2 = \varepsilon_k^2 + |\Delta|^2$,

$$\Delta^* x_\pm + (\varepsilon_k + E_k^\pm)y_\pm \stackrel{(\text{I}')}{=} \Big(\Delta^* + \frac{\varepsilon_k^2 - (E_k^\pm)^2}{\Delta}\Big)x_\pm = 0, \tag{152}$$

i.e. it is satisfied identically. Hence $x_\pm$ can be chosen arbitrarily. $(\odot)$

Normalization.

Requiring $|x_\pm|^2 + |y_\pm|^2 \overset{!}{=} 1$ and using (I′),

$$|x_\pm|^2\Big(1 + \frac{(\varepsilon_k - E_k^\pm)^2}{|\Delta|^2}\Big) = |x_\pm|^2\,\frac{|\Delta|^2 + (\varepsilon_k - E_k^\pm)^2}{|\Delta|^2} \overset{!}{=} 1, \tag{153}$$

where

$$\begin{aligned}(\varepsilon_k - E_k^\pm)^2 &= \varepsilon_k^2 - 2\varepsilon_k E_k^\pm + (E_k^\pm)^2 &(154)\\ &= \varepsilon_k^2 - 2\varepsilon_k E_k^\pm + \varepsilon_k^2 + |\Delta|^2 &(155)\\ &= 2\varepsilon_k^2 - 2\varepsilon_k E_k^\pm + |\Delta|^2. &(156)\end{aligned}$$

Therefore

$$\begin{aligned}|x_\pm|^2 &= \frac{|\Delta|^2}{2\big(\varepsilon_k^2 - \varepsilon_k E_k^\pm + |\Delta|^2\big)} \overset{(*_1)}{=} \frac{1}{2}\,\frac{|\Delta|^2}{(E_k^\pm)^2 - \varepsilon_k E_k^\pm} \overset{(*_1)}{=} \frac{1}{2}\,\frac{(E_k^\pm)^2 - \varepsilon_k^2}{E_k^\pm\big(E_k^\pm - \varepsilon_k\big)} &(157)\\ &= \frac{1}{2}\,\frac{E_k^\pm + \varepsilon_k}{E_k^\pm} = \frac{1}{2}\Big(1 + \frac{\varepsilon_k}{E_k^\pm}\Big) = \frac{1}{2}\Big(1 \pm \frac{\varepsilon_k}{E_k}\Big), &(158)\\ |y_\pm|^2 &= 1 - |x_\pm|^2 = \frac{1}{2}\Big(1 - \frac{\varepsilon_k}{E_k^\pm}\Big) = \frac{1}{2}\Big(1 \mp \frac{\varepsilon_k}{E_k}\Big). &(\mathrm{A})\end{aligned}$$

Fixing the free phase.

Choose $x_\pm$ to be some real value, which is allowed by ($\odot$). Then (A) gives $x_\pm^2 = y_\mp^2$, i.e. $x_\pm = \pm y_\mp$, which is still too many solutions. Set

$$x_+ = y_- \equiv u_k, \qquad x_- = -y_+ \equiv v_k. \tag{159}$$

Since $M_k^\dagger = M_k$, we know that M_k can be diagonalized by a unitary U_k:

$$M = \begin{pmatrix}\varepsilon_k & -\Delta\\ -\Delta^* & -\varepsilon_k\end{pmatrix} = U_k^\dagger \begin{pmatrix}E_k^+ & 0\\ 0 & E_k^-\end{pmatrix} U_k, \qquad U_k = \begin{pmatrix}x^+ & x^-\\ y^+ & y^-\end{pmatrix} = \begin{pmatrix}u_k & v_k\\ -v_k & u_k\end{pmatrix}, \tag{160}$$

which is obviously unitary.

The Bogoliubov transformation.

Inserting this into the Hamiltonian yields

$$\begin{aligned}\tilde{H}_{\text{eff}} &= \sum_k \begin{pmatrix}c_{k\uparrow}^\dagger & c_{-k\downarrow}\end{pmatrix} U_k U_k^\dagger \begin{pmatrix}\varepsilon_k & -\Delta\\ -\Delta^* & -\varepsilon_k\end{pmatrix} U_k U_k^\dagger \begin{pmatrix}c_{k\uparrow}\\ c_{-k\downarrow}^\dagger\end{pmatrix} - \sum_k \varepsilon_k &(161)\\ &= \sum_k \begin{pmatrix}\alpha_k^\dagger & \beta_k\end{pmatrix} \begin{pmatrix}E_k^+ & 0\\ 0 & E_k^-\end{pmatrix} \begin{pmatrix}\alpha_k\\ \beta_k^\dagger\end{pmatrix} - \sum_k \varepsilon_k &(162)\\ &= \sum_k E_k^+ \alpha_k^\dagger \alpha_k + E_k^- \beta_k \beta_k^\dagger - \sum_k \varepsilon_k &(163)\\ &= \sum_k E_k^+\big(\alpha_k^\dagger \alpha_k + \beta_k^\dagger \beta_k\big) - \underbrace{\sum_k \big(\varepsilon_k + E_k^+\big)}_{\text{const.}}, \qquad \sim (5.46) &(164)\end{aligned}$$

where $\{\beta_k, \beta_k^\dagger\} = 1$ and $E_k^- = -E_k^+$ were used, and with $\begin{pmatrix}\alpha_k \\ \beta_k^\dagger\end{pmatrix} = U_k^\dagger \begin{pmatrix}c_{k\uparrow} \\ c_{-k\downarrow}^\dagger\end{pmatrix}$. Here U_k is known as the Bogoliubov transformation, with the corresponding Bogoliubov quasiparticles

$$\alpha_k = u_k c_{k\uparrow} - v_k c_{-k\downarrow}^\dagger, \tag{165}$$

$$\beta_k = u_k c_{-k\downarrow} + v_k c_{k\uparrow}^\dagger, \tag{166}$$

which reproduces (5.43)–(5.47). They obey the usual fermionic anticommutation relations, as can be seen as follows (for arbitrary unitary transformations).

Nambu spinors and the anticommutation relations.

Define

$$\psi_k = \begin{pmatrix}c_{k\uparrow} \\ c_{-k\downarrow}^\dagger\end{pmatrix}, \qquad \psi_k^\dagger = \begin{pmatrix}c_{k\uparrow}^\dagger & c_{-k\downarrow}\end{pmatrix} \qquad \text{(Nambu spinors)}, \tag{167}$$

with $\{\psi_{k,i}, \psi_{k,j}^\dagger\} = \psi_{k,i}\psi_{k,j}^\dagger + \psi_{k,j}^\dagger\psi_{k,i} = \delta_{ij}$. The Bogoliubov transformation can then be written as

$$\Gamma_k \equiv \begin{pmatrix}\alpha_k \\ \beta_k^\dagger\end{pmatrix} = U^\dagger \psi_k, \qquad \Gamma_k^\dagger \equiv \psi_k^\dagger U, \tag{168}$$

so that, using bilinearity of $\{\cdot,\cdot\}$ and that the $U_{\alpha\beta}^{(\dagger)}$ are numbers,

$$\{\Gamma_{k,i}, \Gamma_{k,j}^\dagger\} = \Big\{ \sum_\ell U_{i,\ell}^\dagger \psi_{k,\ell},\ \sum_m \psi_{k,m}^\dagger U_{m,j} \Big\} \tag{169}$$

$$= \sum_{\ell,m} U_{i,\ell}^\dagger \{\psi_{k,\ell}, \psi_{k,m}^\dagger\} U_{m,j} \tag{170}$$

$$= \sum_\ell U_{i,\ell}^\dagger U_{\ell,j} = (U^\dagger U)_{ij} = \delta_{ij}. \tag{171}$$

Since U is unitary, α_k and β_k are fermionic operators.

Coherence factors.

With (5.45),

$$\big(u_k v_k\big)^2 = \frac{1}{4}\Big(1 - \frac{\varepsilon_k^2}{(E_k^+)^2}\Big) = \frac{1}{4}\Big(\frac{\varepsilon_k^2 + |\Delta|^2 - \varepsilon_k^2}{\varepsilon_k^2 + |\Delta|^2}\Big) \tag{172}$$

$$= \frac{1}{4}\,\frac{|\Delta|^2}{(E_k^+)^2} \qquad \Longrightarrow \qquad u_k v_k = \pm\frac{1}{2}\,\frac{|\Delta|}{E_k^+}, \tag{173}$$

from which the rest follows, i.e. (5.51), where the negative solution leads to the contradiction $1 =$ negative value.

The BCS ground state.

Since α_k is fermionic, $n_{\alpha,k}^2 \equiv (\alpha_k^\dagger\alpha_k)^2 = \alpha_k^\dagger\alpha_k$, and thus any eigenvalue λ of $n_{\alpha,k}$ must satisfy $\lambda^2 = \lambda$, i.e. $\lambda \in \{0, 1\}$ (and the same for $n_{\beta,k}$). Since $E_k^+ \geq 0$, it follows that if $\alpha_k|\psi_{\rm BCS}\rangle = \beta_k|\psi_{\rm BCS}\rangle = 0$, then $|\psi_{\rm BCS}\rangle$ is the ground state of $H_{\rm eff} = \sum_k E_k^+(\alpha_k^\dagger\alpha_k + \beta_k^\dagger\beta_k)$.

The pairing ansatz.

Following Becca & Sorella (1.91),

$$|\psi_{\rm BCS}\rangle = \exp\Big(\sum_k f_k c^\dagger_{k\uparrow}c^\dagger_{-k\downarrow}\Big)|0\rangle \tag{174}$$

$$= \prod_k \exp\big(f_k c^\dagger_{k\uparrow}c^\dagger_{-k\downarrow}\big)|0\rangle \tag{175}$$

$$= \prod_k \Big[\sum_{n=0}^{\infty}\frac{f_k^n}{n!}\big(c^\dagger_{k,\uparrow}\big)^n\big(c^\dagger_{-k,\downarrow}\big)^n\Big]|0\rangle, \qquad \text{where } \big(c^\dagger_\alpha\big)^2|0\rangle = 0, \tag{176}$$

$$= \prod_k \big(1 + f_k c^\dagger_{k,\uparrow}c^\dagger_{-k,\downarrow}\big)|0\rangle \tag{177}$$

$$= \bigotimes_k \Big(|0\rangle_k + f_k c^\dagger_{k\uparrow}c^\dagger_{-k\downarrow}|0\rangle_k\Big), \tag{178}$$

with the one-momentum vacuum $|0\rangle_k$ defined by $|0\rangle = \bigotimes_k |0\rangle_k$ and $c^\dagger_{k\uparrow}|0\rangle_k = c^\dagger_{-k\downarrow}|0\rangle_k = 0$. Thus this ansatz is a pure product state, $|\psi_{\rm BCS}\rangle = \bigotimes_k |\psi_{\rm BCS}\rangle_k$, and therefore

$$\alpha_k|\psi_{\rm BCS}\rangle = \alpha_k \bigotimes_{k'} |\psi_{\rm BCS}\rangle_{k'} = |\psi\rangle_{k_1}\otimes\cdots\otimes\alpha_k|\psi_{\rm BCS}\rangle_k\otimes\ldots \tag{179}$$

so that only a single momentum sector has to be examined:

$$\alpha_k|\psi_{\rm BCS}\rangle_k = \big(u_k c_{k\uparrow} - v_k c^\dagger_{-k\downarrow}\big)\Big(|0\rangle_k + f_k c^\dagger_{k\uparrow}c^\dagger_{-k\downarrow}|0\rangle_k\Big) \tag{180}$$

$$= \big(u_k f_k c_{k\uparrow}c^\dagger_{k\uparrow}c^\dagger_{-k\downarrow} - v_k c^\dagger_{-k\downarrow}\big)|0\rangle_k \stackrel{!}{=} 0 \tag{181}$$

$$\iff \quad v_k c^\dagger_{-k\downarrow}|0\rangle_k = u_k f_k\big(1 - c^\dagger_{k\uparrow}c_{k\uparrow}\big)c^\dagger_{-k\downarrow}|0\rangle_k \tag{182}$$

$$= u_k f_k\Big(c^\dagger_{-k\downarrow} + c^\dagger_{k\uparrow}\underbrace{c^\dagger_{-k\downarrow}c_{k\uparrow}}_{=0}\Big)|0\rangle_k, \tag{183}$$

where $c_\alpha|0\rangle = 0$ and $c^\dagger_\alpha c^\dagger_\alpha|0\rangle = 0$ were used. Since $u_k \neq 0$ this gives $f_k = v_k/u_k$, and similarly

$$\beta_k|\psi_{\rm BCS}\rangle_k = \big(u_k f_k c_{-k\downarrow}c^\dagger_{k\uparrow}c^\dagger_{-k\downarrow} + v_k c^\dagger_{k\uparrow}\big)|0\rangle_k \stackrel{!}{=} 0 \tag{184}$$

$$\iff \quad v_k c^\dagger_{k\uparrow}|0\rangle_k = u_k f_k\, c^\dagger_{k\uparrow}\big(1 - c^\dagger_{-k\downarrow}c_{-k\downarrow}\big)|0\rangle_k, \tag{185}$$

again yielding $f_k = v_k/u_k$.

Therefore $|\psi_{\rm BCS}\rangle$ is indeed the ground state of the mean-field Hamiltonian, and it behaves like the vacuum of the Bogoliubons. To avoid problems due to $u_k = 0$, one often writes instead

$$|\psi_{\rm BCS}\rangle = \prod_k \big(u_k + v_k c^\dagger_{k\uparrow}c^\dagger_{-k\downarrow}\big)|0\rangle. \tag{186}$$

3.3 References

- G. Czycholl, *Solid State Theory*, Volume 2 (and Sec. 5.3 of Volume 1 for Kramers' theorem).

- H. Bruus & K. Flensberg, *Many-Body Quantum Theory in Condensed Matter Physics*, Sec. 3.8.
- F. Becca & S. Sorella, *Quantum Monte Carlo Approaches for Correlated Systems*, Eq. (1.91).
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4 Computing similarity transformations of operators

4.1 Motivation

Similarity transformations $B = P^{-1}AP$ are ubiquitous whenever a Hamiltonian is brought to a simpler (often diagonal) form. Common examples include the Bogoliubov transformations of quadratic Hamiltonians, displacement/squeezing of bosonic modes, polaron (Lang-Firsov) transformations, Schrieffer-Wolff transformations, and passage to the interaction picture or a rotating frame are all instances of this construction. Here, we focus on similarity transformation of the form $B \mapsto e^{A}Be^{-A}$ with $A^{\dagger} = -A$ (generator anti-Hermitian). Hence, $U = e^{A}$ is unitary and the transformation preserves Hermiticity and spectra. Similarity transformations are also called equivalence or canoncial transformation and play a big role in group theory.

4.2 Method A: The Hadamard lemma

Hadamard Lemma.
For any two operators A, B,

$$e^{A}Be^{-A} = \sum_{n=0}^{\infty} \frac{1}{n!} \operatorname{ad}_A^n(B), \qquad \text{with} \tag{187}$$

$$\operatorname{ad}_A^n(B) \equiv [A, B]_n \equiv [A, [A, B]_{n-1}], \quad \operatorname{ad}_A^0(B) \equiv [A, B]. \tag{188}$$

This series is especially helpful whenever the nested commutator yields zero at for a finite n or if the nested commutators eventually reproduce a small, recognizable pattern.

Strategy 1: Reduce to elementary operators.

Utilize that for any A, X, Y,

$$e^{A}(XY)e^{-A} = \big(e^{A}Xe^{-A}\big)\big(e^{A}Ye^{-A}\big). \tag{189}$$

It thus suffices to compute $e^{A}\, a\, e^{-A}$ and $e^{A}\, a^{\dagger} e^{-A}$ once and any polynomial in these follows by substitution and normal-ordering, with no further Hadamard series needed.

Strategy 2: The dagger shortcut.

If $A^{\dagger} = -A$, $U = e^{A}$ is unitary with $U^{\dagger} = e^{-A} = U^{-1}$, so $e^{A}Be^{-A} = UBU^{\dagger}$. Therefore, taking the dagger simply yields

$$\big(e^{A}Be^{-A}\big)^{\dagger} = e^{A}B^{\dagger}e^{-A}. \tag{190}$$

Once $e^{A}ae^{-A}$ is known, $e^{A}a^{\dagger}e^{-A}$ follows for free by Hermitian conjugation, roughly halving the bookkeeping.

4.3 Method B: The generator flow method (not proof read since not used yet)

Define $X(\theta) \equiv e^{\theta A}\, X\, e^{-\theta A}$, $X(0) = X$. Differentiating,

$$\frac{dX(\theta)}{d\theta} = [A, X(\theta)] = \mathrm{ad}_A\big(X(\theta)\big). \tag{191}$$

This ordinary differential equation (ODE) is exactly equivalent to Method A – the Hadamard series is its formal Taylor solution around $\theta = 0$ – but is can be more tractable in practice.

Key fact: closure ⇒ finite-dimensional linear ODE.

If ad_A maps a finite set of operators $\{X_1, \ldots, X_n\}$ into linear combinations of themselves, the operator ODE collapses to a matrix ODE:

$$\frac{d}{d\theta}\boldsymbol{X}(\theta) = M\,\boldsymbol{X}(\theta), \qquad [A, X_j] = \sum_i M_{ij} X_i, \qquad \boldsymbol{X}(\theta) = e^{\theta M}\boldsymbol{X}(0), \tag{192}$$

reducing the problem to exponentiating an ordinary $n \times n$ matrix.

4.4 When to prefer which method

Both methods are mathematically identical. Method A is fast when the pattern is recognizable by eye; Method B is more systematic for anything beyond the simplest single-operator case, especially with mixed or multiple generators to track jointly.

4.5 Scope: when this machinery fails

The closure property requires A to be at most quadratic in a set of canonically (anti-) commuting operators. For a genuinely quartic operator such as $H_{\mathrm{int}} = U \sum_i (n_i - \frac{1}{2})(n_{i+1} - \frac{1}{2})$, $\mathrm{ad}_A(H_{\mathrm{int}})$ does not close on a finite set for generic one-body A (e.g. $A \propto J$ or $A \propto T$) – $[J, H_{\mathrm{int}}]$ is generically nonzero and does not reduce to a small closed set of operators. The Hadamard series then cannot be resummed in closed form; one is limited to a perturbative treatment in the strength of A.

4.6 References

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- B.C. Hall, *Lie Groups, Lie Algebras, and Representations.*

5 Group theory

5.1 Basic definitions

Definition 1.
A *representation* of dimension n of the abstract group G is defined as a homomorphism $D : G \to \mathrm{GL}(n, \mathbb{C})$, the group of non-singular $n \times n$ matrices with complex entries.

In other words it is a mapping under which $g \mapsto D(g)$, perserving the group structure:

$$D(g_1 g_2) = D(g_1)D(g_2) \qquad \text{(homomorphism)}. \tag{193}$$

The mapping is necessarily into the set of non-singular matrices, since each matrix must be invertible: $D(g^{-1}) = (D(g))^{-1}$.

Definition 2.
Two elements a and b of a group G are *conjugate* if there exists an element $g \in G$ such that $a = gbg^{-1}$. The element g is called the *conjugating element*. This is an example of an *equivalence relation*[4].

Conjugacy classes are defined as

$$(a) = \{b \mid \exists g \in G : b = gag^{-1}\} \tag{194}$$

and are just one example of *equivalence classes*. The equivalence classes are disjoint sets, that cover the whole set G.

Definition 3.
The character of a representation D of a group G is the set $\chi = \{\chi(g) \mid g \in G\}$, where $\chi(g)$ is the trace of the representation matrix $D(g)$:

$$\chi(g) = \mathrm{Tr}(D(g)). \tag{195}$$

Characters are useful, since they are invariant under equivalence transformations $D(g) \to SD(g)S^{-1}$ due to the cyclic property of the trace.

Definition 4.
A representation of dimension $n + m$ is said to be *reducible* if $D(g)$ takes the form

$$D(g) = \begin{pmatrix} A(g) & C(g) \\ O & B(g) \end{pmatrix} \tag{196}$$

$\forall g \in G$, where A, C and B are submatrices of dimension $m \times m$, $m \times n$ and $n \times n$ respectively, and O denotes a null matrix of dimension $n \times m$.

[4] A relation must be *reflexive*, *symmetric* and *transitive* to be an equivalence relation.

From $D(g)D(g') = D(gg')$ follows

$$A(gg') = A(g)A(g')$$
$$B(gg') = B(g)B(g'),$$

so that $\{A(g)\}$ and $\{B(g)\}$ constitute $[m]$ and $[n]$ representations of G respectively. For finite groups it can be shown (see Maschke's theorem, below) that under equivalence, C can be taken to be a null matrix. The representation is then said to be *completely reducible* or *decomposable*, and we write

$$D(g) = A(g) \oplus B(g).$$

It may be that the representations A and B are themselves decomposable, in which case it is natural to continue the process, which will terminate when we reach the level of *irreducible representations* (irrep), namely representations which cannot be further reduced. Thus, the matrix representation D of the abstract group G can be written as a direct sum of irreducible representations.

Definition 5.
A map $T : V \to V$ is *linear* if

$$T(\alpha \boldsymbol{u} + \beta \boldsymbol{v}) = \alpha(T\boldsymbol{u}) + \beta(T\boldsymbol{v}) \tag{197}$$

for all $\boldsymbol{u}, \boldsymbol{v} \in V; \alpha, \beta \in \mathbb{C}$.

G-module

It turns out to be extremely useful to think of a representation of a group G as a homomorphism, mapping G into the group of non-singular[5] linear transformations of a vector space V. That is, every element g of G is mapped into a linear transformation $T(g)$ with the all-important property that

$$T(g'g) = T(g')T(g) \qquad \text{(homomorphism)}. \tag{198}$$

A vector space in which a group G acts in this way is called a *G-module*.

Reducibility

In this language reducibility corresponds to the existence of a *submodule*, i.e. a subspace U of the vector space V which itself is closed under the action of the group:

$$\boldsymbol{u} \in U \subset V \Rightarrow T(g)\boldsymbol{u} \in U. \tag{199}$$

Using some basis of V to express $T(g)$ as a matrix $D(g)$, one can easily show that the aforementioned closure of U is equivalent to $D(g)$ having the form as in eq. (196).

[5]Non-singular means, that $T(\boldsymbol{v}) = 0 \Rightarrow \boldsymbol{v} = \boldsymbol{0}$, which is equivalent to T being invertible.

Unitary transformations

A linear transformation T acting in the vector space V is said to be *unitary* if

$$(T\boldsymbol{u}, T\boldsymbol{v}) = (\boldsymbol{u}, \boldsymbol{v}) \qquad \forall \boldsymbol{u}, \boldsymbol{v} \in V, \tag{200}$$

where $(\cdot\,, \cdot)$ denotes the scalar product. This is equivalent to

$$(T\boldsymbol{u}, \boldsymbol{v}) = (\boldsymbol{u}, T^{-1}\boldsymbol{v}), \tag{201}$$

since T is invertible.

Complete reducibility

Again, let U be a submodule of the G-module V, closed under the action of the group: $\boldsymbol{u} \in U \subset V \Rightarrow T(g)\boldsymbol{u} \in U$. Use the scalar product to obtain an orthonormal basis of U and of $W = V \setminus U$ such that W is the orthogonal complement of U:

$$W = \{\boldsymbol{w} \in V \mid (\boldsymbol{w}, \boldsymbol{u}) = 0 \quad \forall \boldsymbol{u} \in U\}. \tag{202}$$

In coordinate-free language, complete reducibility corresponds to the closure of W as well as U under the aciton of the group. This will always be the case if the $T(g)$ are unitary with respect to the scalar product we are using. For then, from eq. (201),

$$(T(g)\boldsymbol{w}, \boldsymbol{u}) = (\boldsymbol{w}, T^{-1}(g)\boldsymbol{u}) = (\boldsymbol{w}, T(g^{-1})\boldsymbol{u}) = (\boldsymbol{w}, \boldsymbol{u}') = 0,$$

which shows that W is closed under the action of the group, as required. This translates to the submatrix $C(g)$ being the null matrix.

Maschke's theorem

All reducible representations of a finite group are completely reducible (decomposable). Since we already know, that all *unitary* representations are decomposable, one only has to show that for a finite group it is always possible to choose a scalar product which makes the representation unitary. This then implies that every reducible matrix representation D is equivalent to a reducible unitary representation D'.

Schur's first lemma

In matrix form this states that any matrix which commutes with all the matrices of an irreducible representation must be a multiple of the unit matrix, i.e.

$$BD(g) = D(g)B \quad \forall g \in G \Rightarrow B = \lambda \mathbb{1}, \tag{203}$$

with $\lambda \in \mathbb{C}$. As a statement about linear operators it says that

$$\hat{B}T(g) = T(g)\hat{B} \quad \forall g \in G \Rightarrow \hat{B} = \lambda E, \tag{204}$$

where $\hat{B} : U \to U$ is the linear operator corresponding to B, and E is the identity map $\boldsymbol{u} \mapsto \boldsymbol{u}$.

Schur's second lemma

This concerns two inequivalent irreducible representations D, D', and in matrix form states that

$$BD(g) = D'(g)B \quad \forall g \in G \Rightarrow B = 0. \tag{205}$$

The corresponding linear operator $\hat{B}$ will in this case be a map $\hat{B} : U \to U'$ between the two vector spaces U and U' supporting the two representations, and the coordinate-free version of eq. (205) is

$$\hat{B}T(g) = T'(g)\hat{B} \quad \forall g \in G \Rightarrow \hat{B} = \hat{O}, \tag{206}$$

where $\hat{O}$ is the linear operator which maps every vector in U onto the null vector in U': $\hat{O}\boldsymbol{u} = \mathbf{0}'$ for all $\boldsymbol{u} \in U$.

The fundamental orthogonality theorem

Let U_ν and U_μ be two G-modules carrying inequivalent irreducible representations for a finite group G. For the matrix representations then holds

$$\sum_g D^{(\mu)}_{ir}(g) D^{(\nu)}_{sj}(g^{-1}) = \frac{[g]}{n_\mu} \delta^{\mu\nu} \delta_{ij} \delta_{rs}. \tag{207}$$

Orthogonality of characters

Recall that the character of a representation D is the set $\{\chi(g)\}$, where $\chi(g)$ is the trace of the matrix $D(g)$. These traces have the following properties:

1. χ is the same for equivalent representations, for which $D'(g) = SD(g)S^{-1}$.
2. χ is the same for conjugate elements, since $D(hgh^{-1}) = D(h)D(g)(D(h))^{-1}$.
3. If D is unitary, i.e. $D^{-1} = D^\dagger$, then

$$\chi(g^{-1}) = \mathrm{Tr}((D(g))^{-1}) = \mathrm{Tr}(D(g)^\dagger) = \chi^*(g), \tag{208}$$

   which is true for all finite groups.

In terms of characters the orthogonality theorem reads

$$\frac{1}{[g]} \sum_g \chi^{(\mu)}(g) \chi^{(\nu)^*}(g) = \delta^{\mu\nu}. \tag{209}$$

Since the characters of conjugate elements are equal, all elements of a conjugacy class will have the same character. Let χ_i denote the distinct characters where $i = 1, \ldots, k$ corresponds to the k conjugacy classes K_i. Equation (209) can thus be rewritten as

$$\frac{1}{[g]} \sum_i k_i \chi^{(\mu)}_i(g) \chi^{(\nu)^*}_i(g) = \delta^{\mu\nu}, \tag{210}$$

with k_i denoting the number of elements in the conjugacy class K_i.

One can prove that the number of inequivalent irreducible representations r is equal to the number of conjugacy classes k.

Decomposition of reducible representations

For a finite group any reducible representation is completely reducible into a direct sum of irreps (Maschke's theorem), which means that the representation matrices can be cast in block diagonal form, the non-zero diagonal blocks being the matrices of irreducible representations. A given irrep may appear more than once in this decomposition: for example, a 5-dimensional representation could decompose into the trivial representation and two copies of the same 2-dimensional irrep. In the general case the decomposition is written

$$D = \bigoplus_{\nu} a_\nu D^{(\nu)}, \tag{211}$$

where the non-negative integer a_ν denotes the number of times the irrep $D^{(\nu)}$ appears in the decomposition. One can show that

$$a_\mu = \frac{1}{[g]} \sum_g \chi(g)\chi^{(\mu)}(g^{-1}) \tag{212}$$

$$= \frac{1}{[g]} \sum_g \chi(g)\chi^{(\mu)^*}(g). \tag{213}$$

Construction of the character table

The characters of the irreducible representations of a finite group are conveniently presented in the form of a table, the rows of which correspond to the different irreps, and the columns to the conjugacy classes of the group. In the construction of this table the principal tools we use are:

1. number of irreps = number of conjugacy classes: $r = k$;
2. $\sum_\mu n_\mu^2 = [g]$ (n_μ is the dimension of the irrep);
3. orthogonality: $\sum_i k_i \chi_i^{(\mu)} \chi_i^{(\mu)^*}$;
4. any other information we can come by (!).

In the last category is the fact that for one-dimensional representations the characters are the same thing as the matrices, and hence the characters themselves must mimic the group multiplication properties.

Abelian groups

All irreducible representations of a finite Abelian group G are one-dimensional.

Take some irreducible matrix representation $D^{(\sigma)}$ of the group G[6]. The matrix $D^{(\sigma)}(g)$ commutes with all the matrices of the representation $D^{(\sigma)}$. Therefore, due to Schur's first lemma, the matrix must be a multiple of the identity: $D^{(\sigma)}(g) = \lambda_g \mathbb{1}$. Thus, all the matrices $D^{(\sigma)}(g)$ are diagonal and would therefore be reducible if the dimension is greater than one, which would be a contradiction to our assumption that $D^{(\sigma)}$ is an irrep.

[6]Each finite group has at least the trivial representation $\rho : G \to \mathrm{GL}(1, \mathbb{C})$, with $\rho(g) = 1$ for all $g \in G$.

5.2 Translation operator in quantum mechanics

Wiki:

The translation operator $\hat{T}(\boldsymbol{x})$ moves particles and fields by the amount $\boldsymbol{x}$ (e.g. $\boldsymbol{x} \in \mathbb{R}^3$). Therefore, if a particle is in an eigenstate $|r\rangle$ of the position operator, then after $\hat{T}(\boldsymbol{x}) = \int d\boldsymbol{r}\, |\boldsymbol{r}+\boldsymbol{x}\rangle \langle \boldsymbol{r}|$ acts on it, the particle is at the position $\boldsymbol{r}+\boldsymbol{x}$:

$$\hat{T}(\boldsymbol{x})\, |\boldsymbol{r}\rangle = |\boldsymbol{r}+\boldsymbol{x}\rangle \tag{214}$$

The set $\mathcal{T}$ of translation operators $\hat{T}(\boldsymbol{x})$ for all $\boldsymbol{x}$, with the operation of multiplication defined as the result of successive translations, satisfies all the axioms of a group. In order to see this, let $\boldsymbol{x}, \boldsymbol{x}', \boldsymbol{r} \in \mathbb{R}^3$. Then

$$\hat{T}(\boldsymbol{x})\hat{T}(\boldsymbol{x}')\, |\boldsymbol{r}\rangle = \hat{T}(\boldsymbol{x})\, |\boldsymbol{r}+\boldsymbol{x}'\rangle = |(\boldsymbol{r}+\boldsymbol{x}')+\boldsymbol{x}\rangle = \hat{T}(\boldsymbol{x}+\boldsymbol{x}')\, |\boldsymbol{r}\rangle\,,$$

from which directly follows closure under the action of the group. The existence of an identity and of an inverse for each element, and associativity follows directly from the fact, that $\mathbb{R}^3$ with addition is a group itself.

6 The tensor product and the Kronecker product as one of its possible representations

6.1 The tensor product of vector spaces

Let $\underline{v} = \{v_i \mid i \in I\}$ and $\underline{w} = \{w_j \mid j \in J\}$ be bases for V and W. Then $V \otimes W$ is the product space, where there exists a basis which can be identified uniquely with the pairs of the cartesian product

$$\underline{v} \times \underline{w} = \big\{(v_i, w_j) \mid i \in I,\ j \in J\big\}, \tag{215}$$

so that

$$\dim(V \otimes W) = \dim V \cdot \dim W. \tag{216}$$

Since $V \otimes W$ is a vector space, an arbitrary element of it can be written as

$$\sum_{(i,j)\in I\times J} c_{ij}\,(v_i \otimes w_j). \tag{217}$$

The bilinear map $\otimes$ is defined by

$$\otimes\colon V \times W \longrightarrow V \otimes W, \tag{218}$$

$$(v, w) \longmapsto v \otimes w = \sum_{(i,j)\in I\times J} a_i b_j\,(v_i \otimes w_j), \tag{219}$$

for $v = \sum_i a_i v_i$ and $w = \sum_j b_j w_j$. As a check of bilinearity, writing $v' = \sum_i a'_i v_i$ and $v'' = \sum_i a''_i v_i$,

$$\begin{aligned}(v' + v'') \otimes w &= \Big(\textstyle\sum_i a'_i v_i + \sum_i a''_i v_i\Big) \otimes w = \Big(\sum_i (a'_i + a''_i)\, v_i\Big) \otimes w \\ &= \sum_{i,j} (a'_i + a''_i)\, b_j\,(v_i \otimes w_j) \\ &= \sum_{i,j} a'_i b_j\,(v_i \otimes w_j) + \sum_{i,j} a''_i b_j\,(v_i \otimes w_j) \\ &= v' \otimes w + v'' \otimes w,\end{aligned} \tag{220}$$

and analogously in the second argument.

6.2 The tensor product of operators

Introduce operators $A\colon V \to V$ and $B\colon W \to W$ with V, W vector spaces. How is $A \otimes B\colon V \otimes W \to V \otimes W$ given? Define a map $\beta\colon V \times W \to V \otimes W$, $\beta(v, w) := Av \otimes Bw$. It is bilinear, e.g.

$$\begin{aligned}\beta(v_1 + v_2, w) &= A(v_1 + v_2) \otimes Bw = (Av_1 + Av_2) \otimes Bw \\ &\overset{\otimes \text{ bilin.}}{=} Av_1 \otimes Bw + Av_2 \otimes Bw = \beta(v_1, w) + \beta(v_2, w),\end{aligned} \tag{221}$$

and analogously in the second argument. By the universal property there exists a unique linear $\tilde{\beta}$ with $\tilde{\beta}(v \otimes w) := \beta(v, w)$, so that

$$(A \otimes B)(v \otimes w) = Av \otimes Bw. \tag{222}$$

6.3 The Kronecker product as a matrix representation

Let $V = K^m$ and $W = K^n$ and identify (this is *not* a unique representation!) $K^m \times K^n \to K^{(m\cdot n)}$ with the Kronecker product, i.e. for

$$v = \begin{pmatrix} a_1 \\ \vdots \\ a_m \end{pmatrix} \quad \& \quad w = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} \qquad \text{set} \qquad v \otimes w := \begin{pmatrix} a_1\, w \\ \vdots \\ a_m\, w \end{pmatrix} = \begin{pmatrix} a_1 b_1 \\ \vdots \\ a_1 b_n \\ \vdots \\ a_m b_1 \\ \vdots \\ a_m b_n \end{pmatrix}. \tag{223}$$

Now choose $\{v_i\}$ to be the standard basis, so that $Av_i = A_{\cdot i}$ (the i-th column of A) and likewise for B. Then

$$\begin{aligned} (A \otimes B)(v_i \otimes v_j) = (Av_i) \otimes (Bv_j) &\overset{\text{std. basis}}{=} A_{\cdot i} \otimes B_{\cdot j} \\ = \begin{pmatrix} A_{1i}\, B_{\cdot j} \\ \vdots \\ A_{mi}\, B_{\cdot j} \end{pmatrix} &= \begin{pmatrix} \begin{pmatrix} A_{1i}B_{1j} \\ \vdots \\ A_{1i}B_{nj} \end{pmatrix} \\ \vdots \\ \begin{pmatrix} A_{mi}B_{1j} \\ \vdots \\ A_{mi}B_{nj} \end{pmatrix} \end{pmatrix}. \end{aligned} \tag{224}$$

Setting $v_i \otimes v_j \equiv w_k$ with $k = j + n\, i$ (for zero-based i), the matrix of $A \otimes B$ is the block matrix

$$\begin{aligned} A \otimes B &= \begin{pmatrix} A_{11}B_{\cdot 1} & \cdots & A_{11}B_{\cdot n} & \cdots & A_{1m}B_{\cdot 1} & \cdots & A_{1m}B_{\cdot n} \\ \vdots & & & & & & \vdots \\ A_{m1}B_{\cdot 1} & \cdots & A_{m1}B_{\cdot n} & \cdots & A_{mm}B_{\cdot 1} & \cdots & A_{mm}B_{\cdot n} \end{pmatrix} \\ &= \begin{pmatrix} A_{11}B & \cdots & A_{1m}B \\ \vdots & \ddots & \vdots \\ A_{m1}B & \cdots & A_{mm}B \end{pmatrix}, \end{aligned} \tag{225}$$

which is the Kronecker product for matrices.

6.4 References

- *Tensorprodukt* and *Kronecker-Produkt*, German Wikipedia.

7 Introduction to the density and the reduced density matrix

7.1 The density operator

Let $\{p_i, |\psi_i\rangle\}$ be an ensemble of pure states, describing a system that is in one of the states $|\psi_i\rangle$ with probability p_i, i.e. the system is not fully known. Define the density operator as

$$\rho = \sum_i p_i \, |\psi_i\rangle\langle\psi_i| \,, \qquad p_i \in \mathbb{R} \;\; \& \;\; \sum_i p_i = 1, \tag{226}$$

which is a positive semi-definite operator of trace one (see for example Nielsen & Chuang).

7.2 Observables on a composite system

Take a system A with some observable M, and a system B, giving the composite system AB (see Nielsen & Chuang Box 2.6). Assume we can measure M on system A; then a corresponding measurement on AB is $\tilde{M} = M \otimes \mathbb{1}_B$.

Proof. Take AB to be prepared in $|m\rangle\,|\psi\rangle$ with $M\,|m\rangle = m\,|m\rangle$. Applying the measurement to $|m\rangle\,|\psi\rangle$ should yield m with probability one, i.e. the corresponding projector for $\tilde{M}$ is $P_m \otimes \mathbb{1}_B$. By the spectral theorem, $\tilde{M} = \sum_m m\, P_m \otimes \mathbb{1}_B = M \otimes \mathbb{1}_B$ for a general state of AB. □

7.3 The reduced density matrix as a partial trace

Measurement averages[7] must be the same whether computed via the full ρ^{AB} or via some ρ^A associated to system A, i.e.

$$\langle \hat{M} \rangle = \mathrm{tr}\big(M\rho^A\big) \overset{!}{=} \mathrm{tr}\big(\tilde{M}\rho^{AB}\big) = \mathrm{tr}\big((M \otimes \mathbb{1}_B)\rho^{AB}\big). \tag{227}$$

Take $\rho^A = \mathrm{tr}_B(\rho^{AB})$ where

$$\mathrm{tr}_B\big(|a_1\rangle\langle a_2| \otimes |b_1\rangle\langle b_2|\big) \equiv |a_1\rangle\langle a_2| \; \mathrm{tr}\big(|b_1\rangle\langle b_2|\big) \qquad \& \qquad \mathrm{tr}_B(\cdot) \text{ linear}. \tag{228}$$

Write ρ^{AB} by using the eigenbasis of M for system A with $M\,|i\rangle = m_i\,|i\rangle$ and $\langle i|j\rangle = \delta_{ij}$:

$$\rho^{AB} = \sum_\alpha p^\alpha \, |\psi\rangle\langle\psi|^\alpha = \sum_\alpha p^\alpha \sum_{ijk\ell} c^\alpha_{ij} (c^\alpha_{k\ell})^* \; |i\rangle_A \, |j\rangle_B \, \langle\ell|_B \, \langle k|_A \,, \tag{229}$$

$$\rho^A = \sum_\alpha \sum_{ijk\ell} p^\alpha c^\alpha_{ij} (c^\alpha_{k\ell})^* \; |i\rangle\langle k|_A \underbrace{\mathrm{tr}\big(|j\rangle\langle\ell|_B\big)}_{=\delta_{j\ell}} = \sum_{\alpha,ijk} p^\alpha c^\alpha_{ij} (c^\alpha_{kj})^* \; |i\rangle\langle k|_A \,. \tag{230}$$

Therefore

$$\mathrm{tr}\big(M\rho^A\big) = \sum_{\alpha,ijk} p^\alpha c^\alpha_{ij} (c^\alpha_{kj})^* \underbrace{\mathrm{tr}\big(m_i \, |i\rangle\langle k|_A\big)}_{=m_i\delta_{ik}} = \sum_{\alpha,ij} p^\alpha m_i \, c^\alpha_{ij} (c^\alpha_{ij})^*, \tag{231}$$

[7] The expectation value for a mixed state can be written as $\langle \hat{O} \rangle \equiv \mathrm{tr}\big(\rho \hat{O}\big) = \sum_i p_i \, \langle\psi_i| \, \hat{O} \, |\psi_i\rangle$.

and similarly $\mathrm{tr}\big((M\otimes \mathbb{1}_B)\rho^{AB}\big) = \sum_{\alpha,ij} p^\alpha m_i\, c^\alpha_{ij}(c^\alpha_{ij})^*$, which proves the claim. One can also easily show that this is the only map that conserves consistency (see Nielsen & Chuang Box 2.6).

7.4 Reduced density matrix of a pure state

Now examine the reduced density matrix for a pure state,

$$\rho_{AB} = |\psi\rangle\langle\psi| = \sum_{ijk\ell} c_{ij}(c_{k\ell})^*\ |i\rangle_A\,|j\rangle_B\,\langle\ell|_B\,\langle k|_A\,, \tag{232}$$

$$\rho_A = \mathrm{tr}_B(\rho_{AB}) = \sum_{ijk} c_{ij}(c_{kj})^*\ |i\rangle\langle k|_A\,, \tag{233}$$

such that the matrix representation is

$$\boxed{\rho_A = \psi\psi^\dagger}, \tag{234}$$

using that the c_{ij} parametrize the matrix $\psi \in \mathbb{C}^{d_A\times d_B}$, i.e. $|\psi\rangle = \sum_{ij} c_{ij}\,|i\rangle_A\,|j\rangle_B$ with $\sum_{ij}|c_{ij}|^2 = 1$.

Furthermore, eq. (233) results in

$$\begin{aligned}\mathrm{tr}\,\rho_A &= \mathrm{tr}\left[\sum_{ijk} c_{ij}(c_{kj})^*\,|i\rangle\langle k|_A\right]\\ &= \sum_{ijkl} c_{ij}(c_{kj})^*\delta_{il}\delta_{kl}\\ &= \sum_{jl} c_{lj}(c_{lj})^* = 1,\end{aligned}$$

which means that that the eigenvalues of the reduced density matrix sum to one. This can be seen using that any hermitian matrix can be diagonalized and the cylic property of the trace

$$\mathrm{tr}\,\rho_A = \mathrm{tr}[ODO^\dagger] = \mathrm{tr}\,D = \sum_i \lambda_i = 1. \tag{235}$$

7.5 Connection to the SVD

Now use the SVD on ψ:

$$\psi = USV^\dagger \qquad \text{with } U\in\mathbb{C}^{d_A\times d_A},\ V\in\mathbb{C}^{d_B\times d_B} \text{ unitary.} \tag{236}$$

Then, using $V^\dagger V = \mathbb{1}$,

$$\rho_A = USV^\dagger VS^\dagger U^\dagger = USS^\dagger U^\dagger, \qquad SS^\dagger = \mathrm{diag}(s_1^2,\ldots,s_r^2,0,\ldots,0), \tag{237}$$

and write $S^2 = \mathrm{diag}(s_1^2, \ldots, s_r^2)$. Furthermore, since ρ_A is hermitian we can diagonalize it with $\rho_A = ODO^\dagger = US^2U^\dagger$ and $D = \mathrm{diag}(\lambda_1, \ldots)$, so that

$$\boxed{\lambda_i = s_i^2}, \tag{238}$$

i.e. the eigenvalues are given by the squared Schmidt values and hence non-negative. Thus, we can conclude together with eq. (235) that

$$\boxed{0 \leq \lambda_i \leq 1}. \tag{239}$$

7.6 References

- M. Nielsen & I. Chuang, *Quantum Computation and Quantum Information* (density operator, partial trace; Box 2.6).
- U. Schollwöck, *The density-matrix renormalization group in the age of matrix product states*, beginning of Chapter 4 (MPS).

8 Entanglement entropies

Entanglement can be quantified in various different fashions. Here, we want to explore the von Neumann and the Rényi entropies.

8.1 The von Neumann entanglement entropy

The von Neumann entropy is defined as

$$S(\rho_A) \equiv -\operatorname{Tr}\big(\rho_A \ln \rho_A\big) \tag{240}$$

S is zero for two unentangled systems and non-zero for entangled systems.

Claim: $S(\rho_A) \equiv -\operatorname{Tr}\big(\rho_A \ln \rho_A\big) = -\sum_i \lambda_i \ln \lambda_i = -\sum_i s_i^2 \ln s_i^2$.

Proof: Since ρ_A is hermitian, we can use the spectral theorem to write it as $\rho_A = \sum_i \lambda_i P_i$ where λ_i are its eigenvalues and the eigenvectors $|u_i\rangle$ give $P_i = |u_i\rangle\langle u_i|$, since

$$\rho_A = UDU^\dagger = \big(|u_1\rangle, \ldots, |u_n\rangle\big) \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix} \begin{pmatrix} \langle u_1| \\ \vdots \\ \langle u_n| \end{pmatrix}, \qquad \langle u_i|u_j\rangle = \delta_{ij}. \tag{241}$$

For a function f defined on the spectrum of ρ_A, the spectral functional calculus[8] gives

$$f(\rho_A) \equiv \sum_i f(\lambda_i)\, |u_i\rangle\langle u_i| = U \operatorname{diag}\big(f(\lambda_i)\big) U^\dagger. \tag{242}$$

Therefore, one can rewrite the von Neumann entanglement entropy as

$$\begin{aligned} S(\rho_A) &\equiv -\operatorname{Tr}\big(\rho_A \ln \rho_A\big) && (243)\\ &= -\operatorname{Tr}\big(\rho_A\, U \ln(\operatorname{diag}(\lambda_i))\, U^\dagger\big) \\ &= -\operatorname{Tr}\big(U \operatorname{diag}(\lambda_i) \underbrace{U^\dagger U}_{=\mathbb{1}} \ln(\operatorname{diag}(\lambda_i)) U^\dagger\big) \\ &= -\sum_i \lambda_i \ln \lambda_i = -\sum_i s_i^2 \ln s_i^2. && (244) \end{aligned}$$

■

Example

In our case, for example, $L = 2$, $N_f = 1$, $N_{\text{ph}} = 1$:

$$\begin{aligned} &\{|0\rangle, |1\rangle\} \;\&\; \{|01\rangle, |10\rangle\} \quad \text{as bases of } \mathcal{H}_b \;\&\; \mathcal{H}_f, && (245)\\ &\Rightarrow \{|0,01\rangle;\ |0,10\rangle;\ |1,01\rangle;\ |1,10\rangle\} \quad \text{as basis of } \mathcal{H}_b \otimes \mathcal{H}_f, && (246)\\ &\Rightarrow |\psi\rangle = \sum_{i=1}^{2}\sum_{j=1}^{2} \psi_{ij}\, |i\rangle_b\, |j\rangle_f \quad \text{with } (\psi)_{ij} = \psi_{ij} \;\&\; \psi \in \mathbb{C}^{2\times 2}, && (247)\\ &\Rightarrow S(\rho_A) = -\sum_\ell s_\ell^2 \ln s_\ell^2. && (248) \end{aligned}$$

[8] See for example Nicholas, J. H. (2008). *Functions of matrices: theory and computation.*

Note that since $\operatorname{tr}\rho^{AB} = \operatorname{tr}\rho^{A} = \operatorname{tr}\rho^{B} = 1$ we must have $\sum_\ell \lambda_\ell = \sum_\ell s_\ell^2 = 1$.

8.2 Direct computation using the SVD

The entanglement entropy for the ground state can be computed directly as follows:

1. Obtain an approximation of the ground state $|\psi_0\rangle = \sum_\alpha c_\alpha |\alpha\rangle \in \mathcal{H}^{AB} = \mathcal{H}^A \otimes \mathcal{H}^B$ as some parametrized variational state $\psi_\theta(\alpha)$.
2. Use the variational state (e.g. $\psi_\theta(\alpha) = c_\alpha$) to evaluate the amplitudes on every allowed configuration α and normalize the resulting coefficient vector.
3. Reshape the coefficients into the matrix $\psi \in \mathbb{C}^{d_A \times d_B}$, respecting the ordering of the product basis.
4. Compute the SVD of ψ to obtain the Schmidt values s_i, yielding the entropy $S = -\sum_i s_i^2 \ln s_i^2$

This approach is also straightforward with exact diagonalization since the ground state is represented by a vector consisting of the desired amplitudes. However, this quickly becomes unfeasible for ED and variational states as the Hilbert space grows exponentially with the system size.

8.3 Computation of the von Neumann entropy in VMC

8.3.1 Constructing the reduced density matrix

Given that $d_A \ll d_B$, one can estimate the small reduced density matrix ρ_A and diagonalize it directly. For illustrative purposes, let subsystem A describe a single photon mode and let subsystem B describe a fermionic system.

Let $n = 0, \ldots, N_{\text{ph}}$ denote the photon occupation and σ a fermionic occupation configuration. Define the wavefunction amplitudes by

$$\psi(n, \sigma) \equiv \langle n, \sigma | \psi \rangle \,. \tag{249}$$

For a normalized state, the matrix elements of ρ_A follow from eqs. (233) and (234)

$$(\rho_A)_{nn'} = \sum_\sigma \psi(n, \sigma)\psi^*(n', \sigma) \tag{250}$$

Since variational states do not necessarily provide normalized amplitudes, we might have to normalize them in the above formula. Defining

$$Z \equiv \sum_{m,\sigma} |\psi(m, \sigma)|^2, \tag{251}$$

we can write the normalized reduced density matrix more generally as

$$\boxed{(\rho_A)_{nn'} = \frac{\sum_\sigma \psi(n, \sigma)\psi^*(n', \sigma)}{Z}}. \tag{252}$$

For a normalized state, $Z = 1$.

8.3.2 A Monte Carlo estimator for the reduced density matrix

For each fermionic configuration σ, define

$$u_n(\sigma) \equiv \frac{\psi(n,\sigma)}{\sqrt{\sum_{m=0}^{N_{\rm ph}} |\psi(m,\sigma)|^2}}, \qquad \boldsymbol{u}(\sigma) = \begin{pmatrix} u_0(\sigma) \\ \vdots \\ u_{N_{\rm ph}}(\sigma) \end{pmatrix} \tag{253}$$

only if the denominator is non-zero. This is almost always the case when using Monte Carlo sampling since these configurations will never be visited, unless you start in one of them. This vector is normalized for every σ with non-zero marginal probability:

$$\boldsymbol{u}(\sigma)^\dagger \boldsymbol{u}(\sigma) = \sum_{n=0}^{N_{\rm ph}} |u_n(\sigma)|^2 = 1. \tag{254}$$

The matrix elements of its outer product are

$$\big(\boldsymbol{u}(\sigma)\boldsymbol{u}(\sigma)^\dagger\big)_{nn'} = \frac{\psi(n,\sigma)\psi^*(n',\sigma)}{\sum_{m=0}^{N_{\rm ph}} |\psi(m,\sigma)|^2}. \tag{255}$$

Now consider the marginal probability distribution of the fermionic configurations,

$$p(\sigma) = \sum_{m=0}^{N_{\rm ph}} p(m,\sigma) \tag{256}$$

$$\equiv \frac{\sum_{m=0}^{N_{\rm ph}} |\psi(m,\sigma)|^2}{\sum_{m,\sigma'} |\psi(m,\sigma')|^2}. \tag{257}$$

Taking the expectation value with respect to this distribution yields

$$\begin{aligned}
\mathbb{E}_{\sigma\sim p(\sigma)}\big[\boldsymbol{u}(\sigma)\boldsymbol{u}(\sigma)^\dagger\big]_{nn'} &= \sum_\sigma \frac{\psi(n,\sigma)\psi^*(n',\sigma)}{\sum_m |\psi(m,\sigma)|^2}\, p(\sigma) \\
&= \sum_\sigma \frac{\psi(n,\sigma)\psi^*(n',\sigma)}{\sum_m |\psi(m,\sigma)|^2}\, \frac{\sum_m |\psi(m,\sigma)|^2}{\sum_{m,\sigma'} |\psi(m,\sigma')|^2} \\
&= \frac{\sum_\sigma \psi(n,\sigma)\psi^*(n',\sigma)}{\sum_{m,\sigma'} |\psi(m,\sigma')|^2} \\
&= (\rho_A)_{nn'}.
\end{aligned} \tag{258}$$

Therefore,

$$\boxed{\rho_A = \mathbb{E}_{\sigma\sim p(\sigma)}\big[\boldsymbol{u}(\sigma)\boldsymbol{u}(\sigma)^\dagger\big]}. \tag{259}$$

Moreover, since we are only interested in an expectation value over σ, we have

$$\begin{aligned}
\mathbb{E}_{(n,\sigma)\sim p(n,\sigma)}\big[F(\sigma)\big] &= \sum_{n,\sigma} p(n,\sigma) F(\sigma) \\
&= \sum_\sigma p(\sigma) F(\sigma) \\
&= \mathbb{E}_{\sigma\sim p(\sigma)}\big[F(\sigma)\big].
\end{aligned} \tag{260}$$

Thus, we can sample either from $p(\sigma)$ or from $p(n,\sigma)$, depending on what is more convenient.

8.3.3 Computation using variational Monte Carlo

The resulting procedure is:

1. Sample a configuration $(\tilde{n}, \sigma)$ using the VMC sampler, i.e. from the joint distribution

$$p(n, \sigma) = \frac{|\psi(n, \sigma)|^2}{Z}. \tag{261}$$

   Discard $\tilde{n}$ and retain σ. Note, that Z does not need to be known in VMC. The retained configurations follow the marginal distribution $p(\sigma)$ (see eq. (260)).

2. Evaluate all amplitudes $\psi(n, \sigma)$ for $n = 0, \ldots, N_{\text{ph}}$ and construct $\boldsymbol{u}(\sigma)$.

3. Compute $\boldsymbol{u}(\sigma)\boldsymbol{u}(\sigma)^\dagger$. Repeat the preceding steps and average the resulting matrices:

$$\widehat{\rho}_A = \frac{1}{M} \sum_{k=1}^{M} \boldsymbol{u}(\sigma_k)\boldsymbol{u}(\sigma_k)^\dagger \approx \rho_A. \tag{262}$$

   The wide hat indicates that we only obtain an estimate of the true density matrix.

4. Diagonalize $\widehat{\rho}_A$ and use its eigenvalues $\widehat{\lambda}_i$ to estimate the von Neumann entropy:

$$\widehat{S} = -\sum_i \widehat{\lambda}_i \ln \widehat{\lambda}_i. \tag{263}$$

8.4 The Rényi entanglement entropies

The Rényi entropy of order n is defined as

$$S_n(\rho_A) \equiv \frac{1}{1-n} \ln(\text{tr}[\rho_A^n]) \tag{264}$$

$$= \frac{1}{1-n} \ln \left(\sum_{i=1}^{d_A} \lambda_i^n \right) \equiv S_n(\boldsymbol{\lambda}), \tag{265}$$

using again, that any hermitian matrix is diagonalizable and the cylic property of the trace. Keep in mind, that $0 \leq \lambda_i \leq 1$ and $\sum_i \lambda_i = 1$.

The Rényi entropy has the following useful properties, which we will prove subsequently:

- The Rényi entropy of order one reduces to the von Neumann entropy.
- $\lim_{n\to\infty} S_n(\boldsymbol{\lambda}) \equiv S_\infty(\boldsymbol{\lambda}) = -\ln \max_i \lambda_i$ and $S_2(\boldsymbol{\lambda}) \leq 2S_\infty(\boldsymbol{\lambda})$.
- We have $S_n \geq S_m$ for $n < m$ and therefore S_2 is a lower bound for the Von Neumann entropy S_1.

Claim: The Rényi entropy of order one reduces to the von Neumann entropy.

Proof: We can examine the first order by using l'Hôpital's rule, which yields

$$\begin{aligned} S_1(\rho_A) &= \lim_{n\to 1} \frac{\ln(\mathrm{tr}[\rho_A^n])}{1-n} \overset{\mathrm{H}}{=} \lim_{n\to 1} \frac{\left(\ln \sum_{i=1}^{d_A} \lambda_i^n\right)'}{(1-n)'} \\ &= -\lim_{n\to 1} \frac{\sum_{i=1}^{d_A} \lambda_i^n \ln \lambda_i}{\sum_{i=1}^{d_A} \lambda_i^n} = -\frac{\sum_{i=1}^{d_A} \lambda_i \ln \lambda_i}{\sum_{i=1}^{d_A} \lambda_i} \\ &= -\sum_{i=1}^{d_A} \lambda_i \ln \lambda_i. \end{aligned}$$

■

Claim: $\lim_{n\to\infty} S_n(\boldsymbol{\lambda}) \equiv S_\infty(\boldsymbol{\lambda}) = -\ln \max_i \lambda_i$ and $S_2(\boldsymbol{\lambda}) \leq 2S_\infty(\boldsymbol{\lambda})$.

Proof: Order the different eigenvalues of ρ_A such that $\lambda_1 \leq \lambda_2 \leq \ldots \leq \lambda_2$. Furthermore, denote the number of degenerate $\lambda_1 \equiv \lambda_{\max}$ as m. It then follows

$$\begin{aligned} S_\infty(\boldsymbol{\lambda}) &= \lim_{n\to\infty} \frac{1}{1-n} \ln \left(\sum_{i=1}^{d_A} \lambda_i^n \right) \\ &= \lim_{n\to\infty} \frac{1}{1-n} \ln \left(\sum_{i=1}^{m} \lambda_{\max}^n + \sum_{i=m+1}^{d_A} \lambda_i^n \right) \\ &= \lim_{n\to\infty} \frac{1}{1-n} \ln \left(m\lambda_{\max}^n + 0 \right) \\ &= \lim_{n\to\infty} \frac{\ln m + n \ln \lambda_{\max}}{1-n} = -\ln \max_i \lambda_i. \end{aligned}$$

The second inequality quickly follows

$$S_2(\boldsymbol{\lambda}) = -\ln(\sum_{i=1}^{d_A} \lambda_i^n) \leq -\ln(\lambda_{\max}^2) = -2S_\infty(\boldsymbol{\lambda}),$$

since the logarithm decreases monotonically. ■

Claim: $S_n \geq S_m$ for $n < m$.

Proof: Define $z_i(n) = \lambda_i^n / \sum_{j=1}^{d_A} \lambda_j^n$, which is a probability distribution, and $F(n) = \ln \sum_{j=1}^{d_A} \lambda_j^n$. We take n to be a continuous variable and differentiate F with respect to it using again that $\frac{d}{dn}\lambda^n = \lambda^n \ln \lambda$:

$$\begin{aligned} F'(n) &= \frac{\sum_{i=1}^{d_A} \lambda_i^n \ln \lambda_i}{\sum_{j=1}^{d_A} \lambda_j^n} = \sum_i z_i(n) \ln \lambda_i \qquad \text{and} \\ F''(n) &= \sum_i z_i(n) (\ln \lambda_i)^2 - \left(\sum_i z_i(n) \ln \lambda_i \right)^2 \\ &= \mathrm{Var}_{z(n)}(\ln \lambda_i) \geq 0, \end{aligned}$$

which implies, that F' is non-decreasing. Thus, we can follow w.l.o.g for $x \geq n$

$$F(x) - F(n) = \int_n^x F'(s)ds \leq \int_n^x F'(n)ds = F'(n)(x-n),$$

which is equivalent to saying that F lays above all of its tangent, since it is convex. Since $F(1) = \ln \sum_i \lambda_i = \ln 1 = 0$, it follows that

$$F(1) = 0 \leq F(n) + F'(n)(1-n).$$

We can employ this now to yield

$$\frac{dS(n)}{dn} = \frac{d}{dn}\frac{F(n)}{1-n} = \frac{(1-n)F'(n) + F(n)}{(1-n)^2} \leq 0,$$

which proves, that S_n is non-increasing for $n \neq 1$. Finally, employing l'Hôpital's rule yields for $n = 1$ that

$$\lim_{n\to 1} = -\frac{1}{2}F''(1) \leq 0.$$

■

8.5 Computation of the second order Rényi entropy in VMC

All orders of the Rényi entropies can also be exactly determined, if the reduced density matrix of one subpartition is given, by computing its eigenvalues and using eq. (265). The paper *Measuring Renyi Entanglement Entropy in Quantum Monte Carlo Simulations* by Hastings et al. shows how to compute the second order Rényi entropy and we outline their approach below.

The general idea is as follows. One defines an operator, where it can be shown that its expectation value is equal to the trace of the reduced density matrix squared. Then one only needs to show how this operator can actually be measured within VMC.

Let the full Hilbert space be partioned into A, B, such that

$$|\psi\rangle = \sum_{\alpha,\beta} C_{\alpha,\beta} |\alpha\rangle |\beta\rangle$$

and define

$$\begin{aligned} \mathrm{Swap}_A |\psi\rangle \otimes |\psi\rangle &= \mathrm{Swap}_A \sum_{\alpha_1,\beta_1,\alpha_2,\beta_2} C_{\alpha_1,\beta_1} D_{\alpha_2,\beta_2} |\alpha_1\rangle |\beta_1\rangle |\alpha_2\rangle |\beta_2\rangle \\ &\equiv \sum_{\alpha_1,\beta_1,\alpha_2,\beta_2} C_{\alpha_2,\beta_1} D_{\alpha_1,\beta_2} |\alpha_1\rangle |\beta_1\rangle |\alpha_2\rangle |\beta_2\rangle . \end{aligned} \tag{266}$$

The expectation value of this swap operator is given as

$$\begin{aligned}
\langle \mathrm{Swap}_A \rangle &\equiv \langle\psi| \langle\psi| \mathrm{Swap}_A |\psi\rangle |\psi\rangle \\
&= \sum_{\boldsymbol{\alpha},\boldsymbol{\beta},\boldsymbol{\alpha}',\boldsymbol{\beta}'} C^*_{\alpha_1,\beta_1} D^*_{\alpha_2,\beta_2} C_{\alpha_1',\beta_1'} D_{\alpha_2',\beta_2'} \langle\alpha_1\beta_1\alpha_2\beta_2| \mathrm{Swap}_A |\alpha_1'\beta_1'\alpha_2'\beta_2'\rangle \\
&= \sum_{\boldsymbol{\alpha},\boldsymbol{\beta},\boldsymbol{\alpha}',\boldsymbol{\beta}'} C^*_{\alpha_1,\beta_1} D^*_{\alpha_2,\beta_2} C_{\alpha_1',\beta_1'} D_{\alpha_2',\beta_2'} \delta_{\beta_1,\beta_1'}\delta_{\beta_2,\beta_2'}\delta_{\alpha_1,\alpha_2'}\delta_{\alpha_1,\alpha_1'} \\
&= \sum_{\boldsymbol{\alpha},\boldsymbol{\beta}} C_{\alpha_1,\beta_1} C^*_{\alpha_2,\beta_1} D^*_{\alpha_2,\beta_2} D_{\alpha_1,\beta_2}.
\end{aligned}$$

On the other side,

$$\begin{aligned}
\rho_A = \mathrm{tr}_B(\rho) &= \sum_{\boldsymbol{\alpha},\boldsymbol{\beta},\tilde{\beta}} \left\langle \tilde{\beta} \right| C_{\alpha_1,\beta_1} |\alpha_1\beta_1\rangle \langle\alpha_2,\beta_2| C^*_{\alpha_2,\beta_2} \left| \tilde{\beta} \right\rangle \\
&= \sum_{\beta,\alpha,\alpha'} C_{\alpha\beta} C^*_{\alpha'\beta} |\alpha\rangle \langle\alpha'| \qquad \text{such that} \\
(\rho_A)_{\alpha_1,\alpha_2} &= \sum_{\beta} C_{\alpha_1\beta} C^*_{\alpha_2\beta}.
\end{aligned}$$

Thus, we find that

$$\begin{aligned}
\langle \mathrm{Swap}_A \rangle &= \sum_{\alpha_1} \alpha_2 (\rho_A)_{\alpha_1,\alpha_2} (\rho_A)_{\alpha_2,\alpha_1} \\
&= \sum_{\alpha_1} \alpha_2 (\rho_A^2)_{\alpha_1,\alpha_1} = \mathrm{tr}(\rho_A^2).
\end{aligned} \tag{267}$$

Now how does this expectation value look like within VMC? Remember that

$$\begin{aligned}
\langle O \rangle &= \frac{\sum_{\boldsymbol{x}} |\psi(\boldsymbol{x})|^2 \sum_{\boldsymbol{x}'} \langle\boldsymbol{x}| O |\boldsymbol{x}'\rangle \frac{\langle\boldsymbol{x}'|\psi\rangle}{\langle\boldsymbol{x}|\psi\rangle}}{\sum_{\boldsymbol{x}''} |\psi(\boldsymbol{x}'')|^2} \\
&= \sum_{\boldsymbol{x}} p(\boldsymbol{x}) \sum_{\boldsymbol{x}'} \langle\boldsymbol{x}| O |\boldsymbol{x}'\rangle \frac{\langle\boldsymbol{x}'|\psi\rangle}{\langle\boldsymbol{x}|\psi\rangle}.
\end{aligned}$$

Computing this for $O = \mathrm{Swap}_A$, where $|\boldsymbol{x}\rangle \to |\alpha_1\beta_1\alpha_2\beta_2\rangle$ and $|\psi\rangle \to |\psi\rangle \otimes |\psi\rangle$ yields

$$\langle \mathrm{Swap}_A \rangle = \sum_{\boldsymbol{\alpha\beta}} p(\alpha_1,\beta_1,\alpha_2,\beta_2) \frac{\psi(\alpha_1,\beta_2)\psi(\alpha_2,\beta_1)}{\psi(\alpha_1,\beta_1)\psi(\alpha_2,\beta_2)} \tag{268}$$

$$= \left\langle \frac{\psi(\alpha,\beta')\psi(\alpha',\beta)}{\psi(\alpha,\beta)\psi(\alpha',\beta')} \right\rangle, \tag{269}$$

with

$$p(\alpha_1,\beta_1,\alpha_2,\beta_2) = \frac{|\psi(\alpha_1,\beta_1)|^2 |\psi(\alpha_2,\beta_2)|^2}{\sum_{\boldsymbol{\alpha}'\boldsymbol{\beta}'} |\psi(\alpha_1',\beta_1')|^2 |\psi(\alpha_2',\beta_2')|^2}.$$

Thus, we can easily access this within VMC by sampling from the distribution given by $\psi(\alpha,\beta)\psi(\alpha',\beta')$ (see also NetKet documentation).

8.6 References

- M. Nielsen & I. Chuang, *Quantum Computation and Quantum Information* (density operator, partial trace; Box 2.6).
- U. Schollwöck, *The density-matrix renormalization group in the age of matrix product states*, beginning of Chapter 4 (MPS).
- Tang, Yifan, et al. *Deep quantum Monte Carlo approach for polaritonic chemistry.* The Journal of Chemical Physics 163.3 (2025). (Appendix A.1 and A.3)
- M. Hastings, et al. *Measuring Renyi Entanglement Entropy in Quantum Monte Carlo Simulations.* Physical review letters 104.15 (2010): 157201. (equation 4)

9 Quick refresher on spin

Spin can be described with the operators $\hat{S}_x, \hat{S}_y, \hat{S}_z$ that obey the following commutation relations:

$$\boxed{[\hat{S}_j, \hat{S}_k] = i\hbar\,\epsilon_{jk\ell}\hat{S}_\ell} \qquad (\hbar = 1), \tag{270}$$

with $i, j, k = x, y, z = 1, 2, 3$ and using the Einstein summation convention. One can define raising and lowering operators as

$$\boxed{\hat{S}_\pm = \hat{S}_x \pm i\hat{S}_y} \qquad \text{with} \qquad \boxed{[S_-, S_+] = -2S_z}. \tag{271}$$

Proof.

$$\begin{aligned}
[S_-, S_+] &= [S_x - iS_y,\ S_x + iS_y] \\
&= [S_x, S_x] + i[S_x, S_y] - i[S_y, S_x] + [S_y, S_y] \\
&= 2i[S_x, S_y] \overset{(270)}{=} 2i \cdot i\,\epsilon_{12\ell}S_\ell = -2\,\epsilon_{12\ell}S_\ell = -2S_z,
\end{aligned} \tag{272}$$

where the first step used the bilinearity of the commutator and the last step uses $\epsilon_{jk\ell} = 0$ if two indices are equal, leaving $\epsilon_{123}S_3 = S_z$. □

For spin-$\frac{1}{2}$ particles, one can write

$$S_j = \frac{\hbar}{2}\,\sigma_j \qquad \text{with } \sigma_j \text{ the Pauli matrices, } j = x, y, z, \tag{273}$$

and often again $\hbar = 1$.

10 Quick refresher on unitary transformations

Let's say we have some Hamiltonian $\hat{H}$, which we examine in a basis $\{|v_j\rangle\}$ with $j = 1, \ldots, \dim(\mathcal{H})$ and $\hat{H}\colon \mathcal{H} \to \mathcal{H}$. The matrix representation of the operator $\hat{H}$ in this basis is thus given by

$$H_{ij} = \left\langle v_i \middle| \hat{H}\, v_j \right\rangle = \langle v_i |\, \hat{H}\, |v_j\rangle\,. \tag{274}$$

Applying a unitary transformation U (with $U^\dagger U = \mathbb{1} = UU^\dagger$), i.e.

$$|v_j\rangle \to |\tilde{v}_j\rangle = U\,|v_j\rangle \qquad \& \qquad \hat{H} \to \hat{\tilde{H}} = U\hat{H}U^\dagger, \tag{275}$$

yields the matrix elements

$$\tilde{H}_{ij} = \langle \tilde{v}_i |\, \hat{\tilde{H}}\, |\tilde{v}_j\rangle = \langle v_i |\, \underbrace{U^\dagger U}_{=\mathbb{1}}\, \hat{H}\, \underbrace{U^\dagger U}_{=\mathbb{1}}\, |v_j\rangle = \langle v_i |\, \hat{H}\, |v_j\rangle = H_{ij}. \tag{276}$$

Hence the matrix representation of an operator does *not* change under a unitary transformation.

11 Quick refresher on the current in electrodynamics

11.1 Maxwell's equations

Given Maxwell's equations,

$$\boldsymbol{\nabla} \cdot \boldsymbol{E}(\boldsymbol{r}, t) = \frac{1}{\varepsilon_0}\, \rho(\boldsymbol{r}, t) \tag{277}$$

$$\boldsymbol{\nabla} \cdot \boldsymbol{B}(\boldsymbol{r}, t) = 0 \tag{278}$$

$$\boldsymbol{\nabla} \times \boldsymbol{E}(\boldsymbol{r}, t) = -\frac{\partial}{\partial t} \boldsymbol{B}(\boldsymbol{r}, t) \tag{279}$$

$$\boldsymbol{\nabla} \times \boldsymbol{B}(\boldsymbol{r}, t) = \frac{1}{c^2} \frac{\partial}{\partial t} \boldsymbol{E}(\boldsymbol{r}, t) + \frac{1}{\varepsilon_0 c^2}\, \boldsymbol{j}(\boldsymbol{r}, t) \tag{280}$$

11.2 Continuity equation

From $\varepsilon_0\, \partial_t$ (*eq.* (277)), using $\partial_x \partial_t = \partial_t \partial_x$,

$$\frac{\partial}{\partial t} \rho(\boldsymbol{r}, t) = \varepsilon_0\, \boldsymbol{\nabla} \cdot \big(\partial_t \boldsymbol{E}(\boldsymbol{r}, t) \big), \tag{281}$$

and from $\varepsilon_0 c^2\, \boldsymbol{\nabla} \cdot$ (*eq.* (280)),

$$\begin{aligned} \boldsymbol{\nabla} \cdot \boldsymbol{j}(\boldsymbol{r}, t) &= \boldsymbol{\nabla} \cdot \big(\boldsymbol{\nabla} \times \boldsymbol{B}(\boldsymbol{r}, t) \big) - \varepsilon_0\, \boldsymbol{\nabla} \cdot \big(\partial_t \boldsymbol{E}(\boldsymbol{r}, t) \big) \\ &= -\varepsilon_0\, \boldsymbol{\nabla} \cdot \big(\partial_t \boldsymbol{E}(\boldsymbol{r}, t) \big) \qquad (\mathrm{div}(\mathrm{rot}\, \boldsymbol{F}) = 0). \end{aligned} \tag{282}$$

Adding the two gives the continuity equation

$$\frac{\partial}{\partial t} \rho(\boldsymbol{r}, t) + \boldsymbol{\nabla} \cdot \boldsymbol{j}(\boldsymbol{r}, t) = 0. \tag{283}$$

11.3 Point-particle solutions

Solutions of the continuity equation:

$$\rho(\boldsymbol{r}, t) = \sum_\alpha q_\alpha\, \delta\big(\boldsymbol{r} - \boldsymbol{r}_\alpha(t) \big) \tag{284}$$

$$\boldsymbol{j}(\boldsymbol{r}, t) = \sum_\alpha q_\alpha\, \boldsymbol{v}_\alpha(t)\, \delta\big(\boldsymbol{r} - \boldsymbol{r}_\alpha(t) \big) \tag{285}$$

Proof. Note that

$$\begin{aligned} \partial_t\, \delta\big(\boldsymbol{r} - \boldsymbol{r}_\alpha(t) \big) &= \partial_t \Big(\delta\big(r_x - r_x^\alpha(t) \big) \delta\big(r_y - r_y^\alpha(t) \big) \delta\big(r_z - r_z^\alpha(t) \big) \Big) \\ &= -\sum_{i=x,y,z} \dot{r}_i^\alpha(t)\, \partial_{r_i} \delta\big(r_i - r_i^\alpha(t) \big) = -\dot{\boldsymbol{r}}_\alpha(t) \cdot \boldsymbol{\nabla}\, \delta\big(\boldsymbol{r} - \boldsymbol{r}_\alpha(t) \big), \end{aligned} \tag{286}$$

where we used the chain rule in the second line and

$$\boldsymbol{\nabla}\cdot\Big(\boldsymbol{v}_\alpha(t)\,\delta\big(\boldsymbol{r}-\boldsymbol{r}_\alpha(t)\big)\Big)=\sum_{i=x,y,z}\partial_{r_i}\Big(v_i^\alpha(t)\,\delta\big(\boldsymbol{r}-\boldsymbol{r}_\alpha(t)\big)\Big)$$
$$=\sum_{i=x,y,z}v_i^\alpha(t)\,\partial_{r_i}\delta\big(\boldsymbol{r}-\boldsymbol{r}_\alpha(t)\big)=\boldsymbol{v}_\alpha(t)\cdot\boldsymbol{\nabla}\,\delta\big(\boldsymbol{r}-\boldsymbol{r}_\alpha(t)\big). \qquad (287)$$

Therefore

$$\frac{\partial}{\partial t}\rho(\boldsymbol{r},t)=-\sum_\alpha q_\alpha\,\dot{\boldsymbol{r}}_\alpha(t)\cdot\Big(\boldsymbol{\nabla}\,\delta\big(\boldsymbol{r}-\boldsymbol{r}_\alpha(t)\big)\Big)=-\boldsymbol{\nabla}\cdot\boldsymbol{j}(\boldsymbol{r},t), \qquad (288)$$

which is (283). □

Note that these solutions are not unique (one equation for four components).

Intermediate result: the current can be defined by "charge times velocity".

12 Minimizing complex functions with Wirtinger derivatives

12.1 Definition

Define for $z = x + iy \in \mathbb{C}$ with $x, y \in \mathbb{R}$

$$\frac{\partial}{\partial z} \equiv \frac{1}{2}\Big(\frac{\partial}{\partial x} - i\frac{\partial}{\partial y}\Big) \qquad \& \qquad \frac{\partial}{\partial \bar{z}} = \frac{1}{2}\Big(\frac{\partial}{\partial x} + i\frac{\partial}{\partial y}\Big) \tag{289}$$

for functions of one complex variable, and

$$\frac{\partial}{\partial z_j} = \frac{1}{2}\Big(\frac{\partial}{\partial x_j} - i\frac{\partial}{\partial y_j}\Big) \qquad \& \qquad \frac{\partial}{\partial \bar{z}_j} = \frac{1}{2}\Big(\frac{\partial}{\partial x_j} + i\frac{\partial}{\partial y_j}\Big) \tag{290}$$

for functions of multiple complex variables.

12.2 Agreement with the complex derivative

If f is complex differentiable at a point (i.e. the limit $f'(z_0) = \lim_{z\to z_0} \frac{f(z)-f(z_0)}{z-z_0}$ exists), then $\frac{\partial f}{\partial z}$ agrees with the complex derivative $\frac{df}{dz}$.

Proof. For $f(z) = u(z) + iv(z)$ complex differentiable,

$$\begin{aligned} \frac{\partial f}{\partial z} &\overset{(289)}{=} \frac{1}{2}\Big(\frac{\partial f}{\partial x} - i\frac{\partial f}{\partial y}\Big) = \frac{1}{2}\Big(\frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x} - i\frac{\partial u}{\partial y} + \frac{\partial v}{\partial y}\Big) \\ &\overset{(289)}{=} \frac{\partial u}{\partial z} + i\frac{\partial v}{\partial z} \overset{\text{Def.}}{=} \frac{df}{dz}. \end{aligned} \tag{291}$$

□

12.3 The $\bar{z}$-derivative and Cauchy–Riemann

Furthermore,

$$\begin{aligned} 0 \overset{!}{=} \frac{\partial f}{\partial \bar{z}} &= \frac{1}{2}\Big(\frac{\partial f}{\partial x} + i\frac{\partial f}{\partial y}\Big) = \frac{1}{2}\Big(\frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x} + i\frac{\partial u}{\partial y} - \frac{\partial v}{\partial y}\Big) \\ &\iff \quad \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \wedge \quad \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}, \end{aligned} \tag{292}$$

so that $0 = \dfrac{\partial f}{\partial \bar{z}}$ is equivalent to the Cauchy–Riemann equations, i.e. the necessary & sufficient condition for f to be complex differentiable.

12.4 Minimizing a real-valued function

Minimize a real-valued function[9]

$$f\colon \mathbb{C} \to \mathbb{R} \tag{293}$$
$$f(z) = f(x,y), \qquad (z = x + iy) \tag{294}$$

via the condition

$$\nabla f(x,y) = \begin{pmatrix} \partial_x f \\ \partial_y f \end{pmatrix} \stackrel{!}{=} 0. \tag{295}$$

Then

$$\partial_x f \stackrel{!}{=} 0 \stackrel{!}{=} \partial_y f \quad \Longleftrightarrow \quad \frac{\partial f}{\partial \bar{z}} = \frac{1}{2}\Big(\frac{\partial f}{\partial x} + i\frac{\partial f}{\partial y}\Big) \stackrel{!}{=} 0, \tag{296}$$

so one can minimize a real-valued function by imposing $\boxed{\frac{\partial f}{\partial \bar{z}} = 0}$.

12.5 References

- *Wirtinger derivatives*, English Wikipedia.

[9] Minimizing a complex-valued f does not make sense because $\mathbb{C}$ has no order.